CampusReConnect Application

SE-505: Software Project Lab II

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Submission Date: 09-03-2025

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Abstract

This project addresses the challenge of limited interdepartmental research collaboration at the University of Dhaka by developing "CampusReConnect," a unified platform designed to foster research connections among students and faculty. This abstract outlines the project's core objectives, software requirements, and anticipated challenges. Inspired by platforms like ResearchGate, CampusReConnect aims to facilitate academic collaboration, knowledge sharing, and research efficiency. The Software Requirements Specification (SRS) document, developed as a blueprint, details the functional and non-functional requirements, including user profile management, research interest tagging, community creation, and communication tools. The platform is designed to serve undergraduate and postgraduate students, as well as faculty members, by providing a centralized space for identifying potential collaborators, sharing research findings, and engaging in interdisciplinary projects.

The accompanying user manual will provide detailed instructions for navigating and utilizing the platform effectively. Key challenges include ensuring user adoption across diverse academic disciplines, managing data privacy and security, and maintaining system scalability in response to growing user demand. Through a robust development process guided by the SRS, coupled with comprehensive user documentation, CampusReConnect seeks to transform the research landscape at the University of Dhaka, promoting innovation and collaboration.

Introduction

The Software Requirements Specifications (SRS) document outlines the functional and non-functional requirements of the **CampusReConnect** - A Platform for Research Activities Within The University of Dhaka. This application aims to create a unified platform that fosters research connections across the university. The SRS provides a comprehensive overview of the system's functionality, intended audience, and key features to be developed.

Motivation

Many research enthusiasts at the University of Dhaka face challenges due to limited interaction across departments, which hampers interdisciplinary research and innovation. This project is motivated by the need to create a unified platform that fosters research connections across the university. By developing this platform, we aim to promote academic collaboration, knowledge sharing, and research efficiency, potentially transforming how students and faculty engage in research endeavors. The core functionalities are inspired by the “ResearchGate”.

Purpose

The purpose of this SRS is to serve as a blueprint for the development team, guiding them in the design, implementation, and testing phases of the **CampusReConnect** Application. By clearly defining the requirements and expectations of the system, this document ensures that all stakeholders have a common understanding of the project scope and objectives. Additionally, the SRS facilitates communication between stakeholders and developers, enabling effective collaboration throughout the software development lifecycle.

Intended Audience

The intended audience for this SRS includes stakeholders involved in the planning, development, and implementation of the **CampusReConnect** Application. This includes:

Development team: Software engineers, designers, and testers responsible for building and testing the application.

University of Dhaka Students: Undergraduate and postgraduate students interested in research and collaboration, looking to connect with peers from other departments for interdisciplinary projects.
University of Dhaka Faculty: Professors, lecturers, and research supervisors seeking to find and mentor students in their research areas, as well as collaborate with faculty from other departments.

Conclusion

In conclusion, the Software Requirements Specifications document serves as a vital document in the development of the **CampusReConnect** Application. By outlining the system's functionality, purpose, and intended audience, this document provides a foundation for successful project execution. With clear requirements and objectives defined, the development team can proceed with confidence, knowing they have a comprehensive understanding of the project scope and goals. Ultimately, the SRS aims to ensure the successful delivery of a robust and user-friendly estate management system that meets the needs of all stakeholders involved.

Inception

Identifying Stakeholders

Stakeholders are individuals or groups who are actively involved in the project, process, or system and directly interact with it. Their roles and responsibilities are clearly defined, and they are impacted by the system's operations, changes, and performance. Following are the direct stakeholders:

1. Faculty members
2. Students

Quality Function Deployment

Normal Functionality

- **User Registration and Login**: Users can sign up using their institutional email and log in with basic credentials.
- **Profile Creation**: Users can create profiles with basic information like name, department, university, and field of expertise.
- **News Feed**: Display a feed where users see posts relevant to their field of study or expertise.
- **Post Creation**: Users can create and share posts that can be tagged with specific academic subjects or research interests.
- **Search and Filtering**: Basic search functionality to find posts (queries and publications) or users by keywords, tags, or departments.

Expected Functionality

- **Domain-based Content Filtering:** Users can filter their feed to see posts related to their specific domains or interests.
- **Networking:** Users can follow other users to find their activities in the feed.
- **User Profile:** Each user profile will have some description regarding his/her field of interest.
    a. Each profile will have scores based on citation and h-index.
    b. Research lists.
- **Messaging:** Mail message for users to discuss research or academic projects privately.
- **Editing:** Users can edit their personal information and can edit content on their posts and comments.
- **Groups/Communities:** Users can join groups or communities based on their areas of interest (e.g., “AI Researchers,” “Math Enthusiasts”).
- **Notifications:** Users get notified about posts, comments, updates, or messages relevant to their fields of interest.
- **User Verification:** Users are verified based on their institutional email to prevent unauthorized access.

Exciting Functionality

- **Automated Summaries/Abstract:** The platform could automatically generate concise summaries of long research papers or posts, helping users quickly digest important information.
- **Gamification:** Introduce a points system where users can earn points for participating in discussions, answering questions, or helping others in research.

Usage Scenario of CampusReConnect application

User Authentication

Students and faculty members at the University of Dhaka can create an account on the platform by entering their institutional email, name, department, and password. Upon registration, the system sends a confirmation email to verify the user's identity. Once verified, the credentials are securely stored in the database.

Following registration, users can log in by providing their institutional email and password, which are authenticated by the system's database. Users can reset forgotten passwords via the "Forgot Password?" feature. Additionally, users have the ability to change their password for enhanced security. At any point, users can log out of their accounts to ensure privacy.

Profile Management

The Profile Management module allows users to create, update, delete and manage their personal profiles, providing a comprehensive view of their academic background. Students and faculty at the University of Dhaka can input their name, department, expertise, and other personal details to build a professional profile that reflects their academic journey.

The profile also displays key research metrics, such as citation count and h-index, enabling others to quickly assess a user's academic impact. Users have the option to edit their profile at any time, allowing them to keep their information current as they progress in their research careers.

Community

A faculty member can create a community with its name and become the moderator of it. The users can join the community. The communities need the approval of the moderator to add a user. A user can leave the community anytime. The moderator can remove any member and delete any post from the community.

Content Feed

The Content Feed is customized for each user, displaying posts relevant to their department, field of study, and research interests. research interests. Students and faculty members can view a curated stream of content, including recent posts, academic discussions, and research publications related to their areas. The feed prioritizes posts from followed profiles and recently updated content. Users can search other users from the search bar.

Post

The Post module empowers users to share their academic insights and engage with the university's research community. Students and faculty at the University of Dhaka can create posts by entering a title, description, and relevant tags to categorize the content. They can attach documents, such as research papers to provide further context and detail. Additionally, users can leverage the paper summarization feature, which automatically generates concise summaries of research papers to make the content more accessible and easier to digest.

Users can also react to posts and comments with upvotes or downvotes, which can help highlight valuable contributions and generate more visibility for important discussions.

To facilitate engagement, users are notified when someone interacts with their posts or when relevant content based on their interests is posted.

Message

The Mail Messaging module allows students and faculty at the University of Dhaka to engage in communication within the platform. Users can send text messages. All conversations will be stored in the database.

Notification

Users receive notifications when there are replies or comments on posts or questions they have created or are following.

Gamification

Gamification elements, such as points and badges, reward users for activities like posting research, answering questions, and engaging with content. Accumulating points unlocks badges that reflect expertise and involvement, encouraging users to contribute more and stay active.

Use Case Diagram

A Use Case describes the system behavior under various conditions as the system responds to a request from one of its stakeholders. In fact, a use case diagram is a kind of visualization of the system where an end-user has an idea of a specific feature. It simply describes a story using corresponding actors who perform important roles in the story and make the story understandable for the users.

The first step in writing a Use Case is to define the set of “actors” that will be involved in the story. Actors are the different people or systems that use the system or product within the context of the function and behavior that is to be described. Actors represent the roles that people play as system operators. They procedure some information or consume some information. Every user has one or more goals when using the system.

Primary Actor

Primary actors interact directly to achieve the required system function and derive the intended benefit from the system. They work directly with the software. They produce some information and consume some information too.

Secondary Actor

Secondary actors support the system so that primary actors can do their work.They either produce or consume information.

Here is the use case diagram to observe the non-technical view of the System.

Actors:

- Student
- Faculty
- Moderator

Level: 0

Name: CampusReConnect
Primary Actor: Faculty, Students, Moderator
Secondary Actor: SendGrid API, Gemini

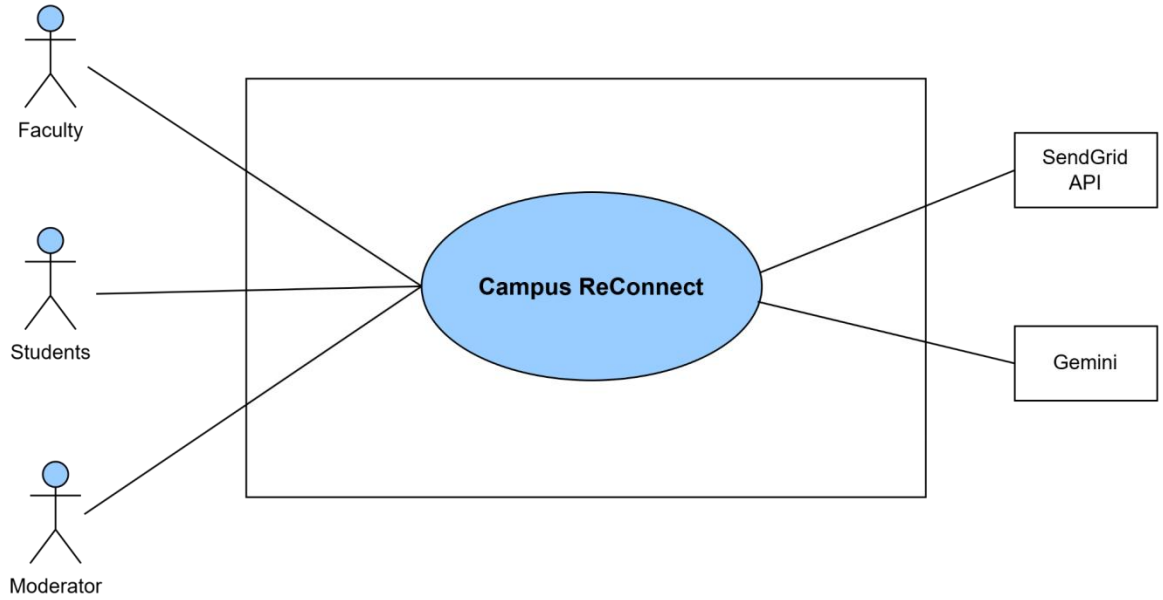


Figure 1. CampusReConnect use case diagram

Level: 1

Name: CampusReConnect(Detailed)
Primary Actor: Faculty, Students, Moderator
Secondary Actor: SendGrid API, Gemini

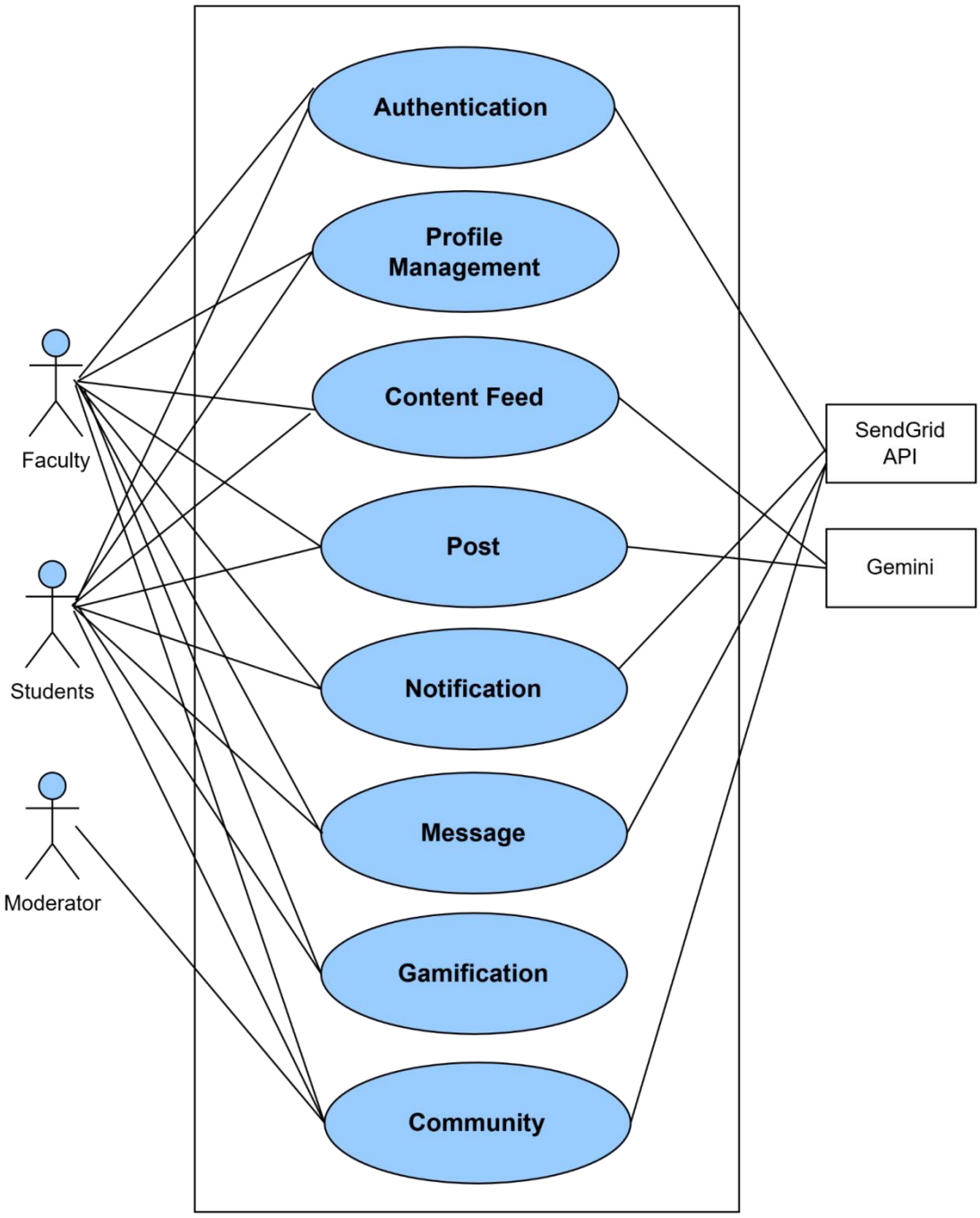


Figure 2. CampusReConnect(Detailed) use case diagram

Level: 1.1

Name: Authentication
Primary Actor: Faculty, Students
Secondary Actor: SendGrid API

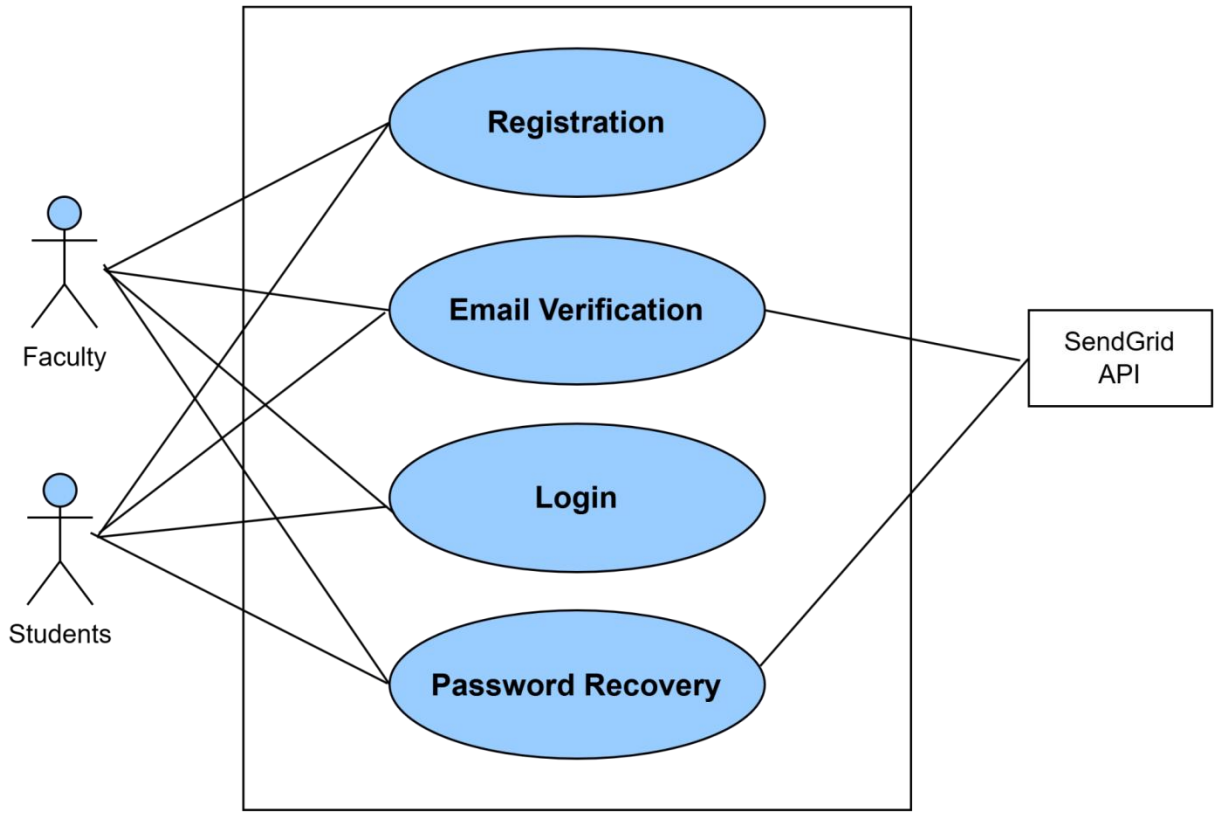


Figure 3. Authentication use case diagram

Level: 1.2

Name: Profile Management
Primary Actor: Faculty, Students

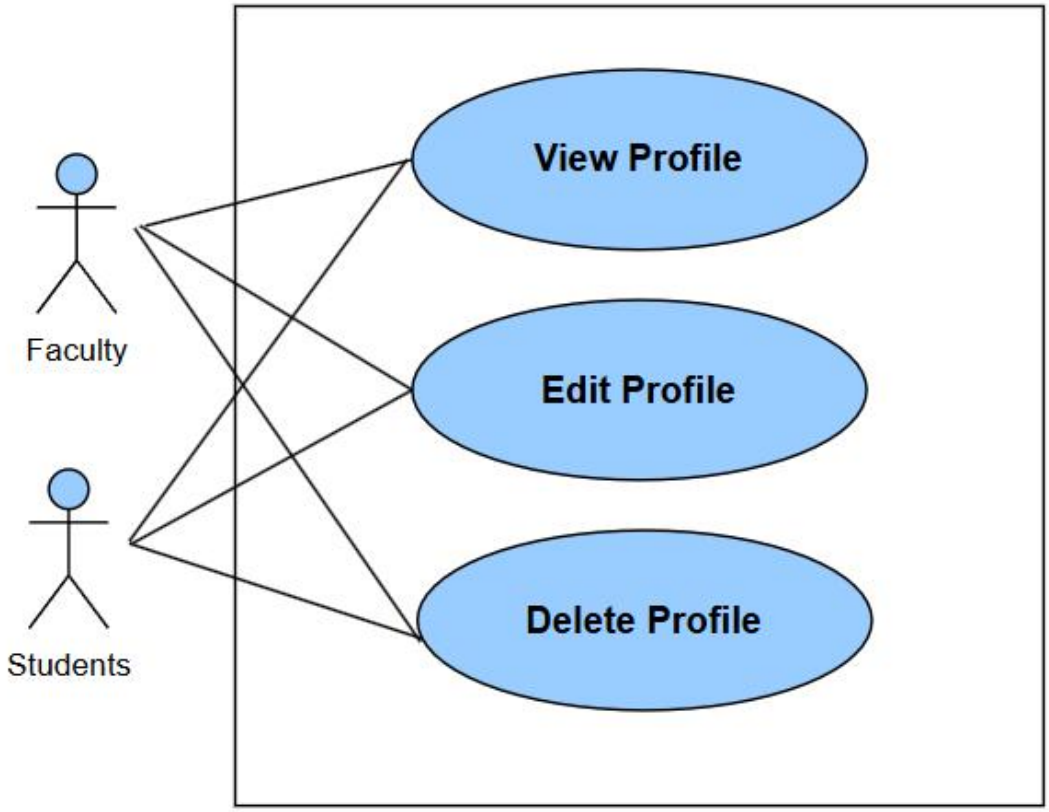


Figure 4. Profile management use case diagram

Level: 1.3

Name: Content Feed
Primary Actor: Faculty, Students

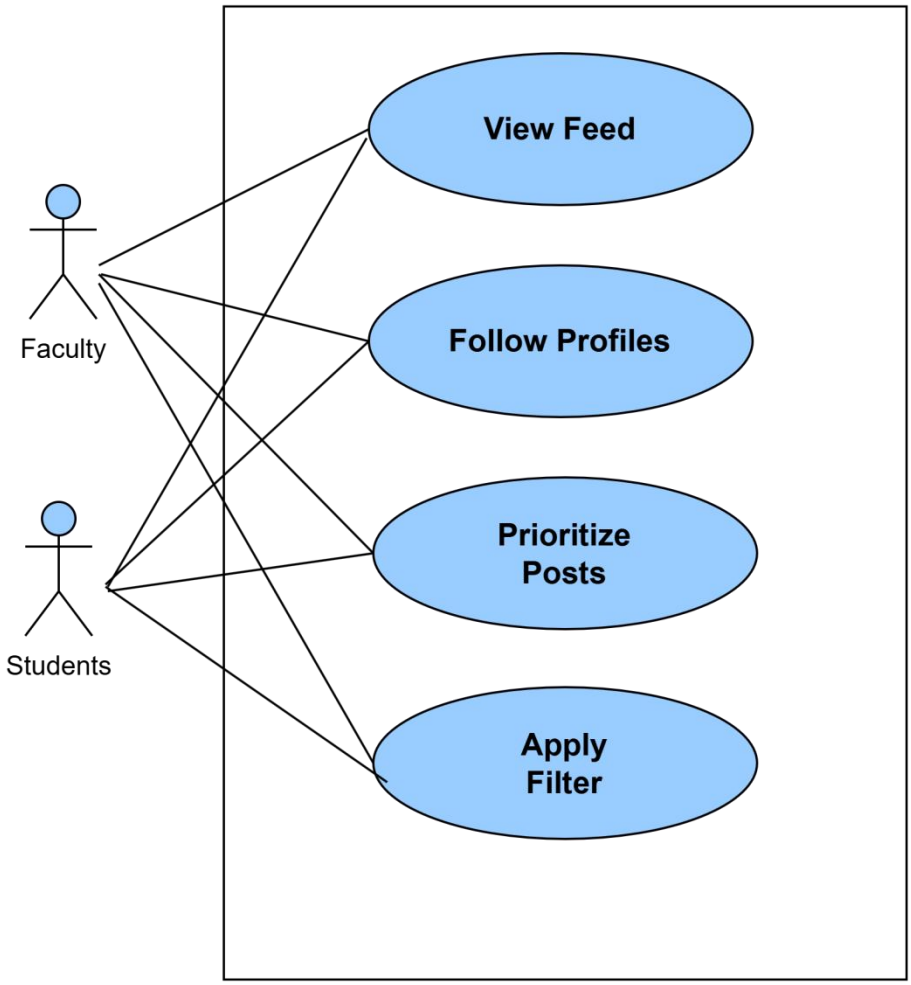


Figure 5. Content feed use case diagram

Level: 1.4

Name: Post
Primary Actor: Faculty, Students
Secondary Actor: Gemini, SendGrid API

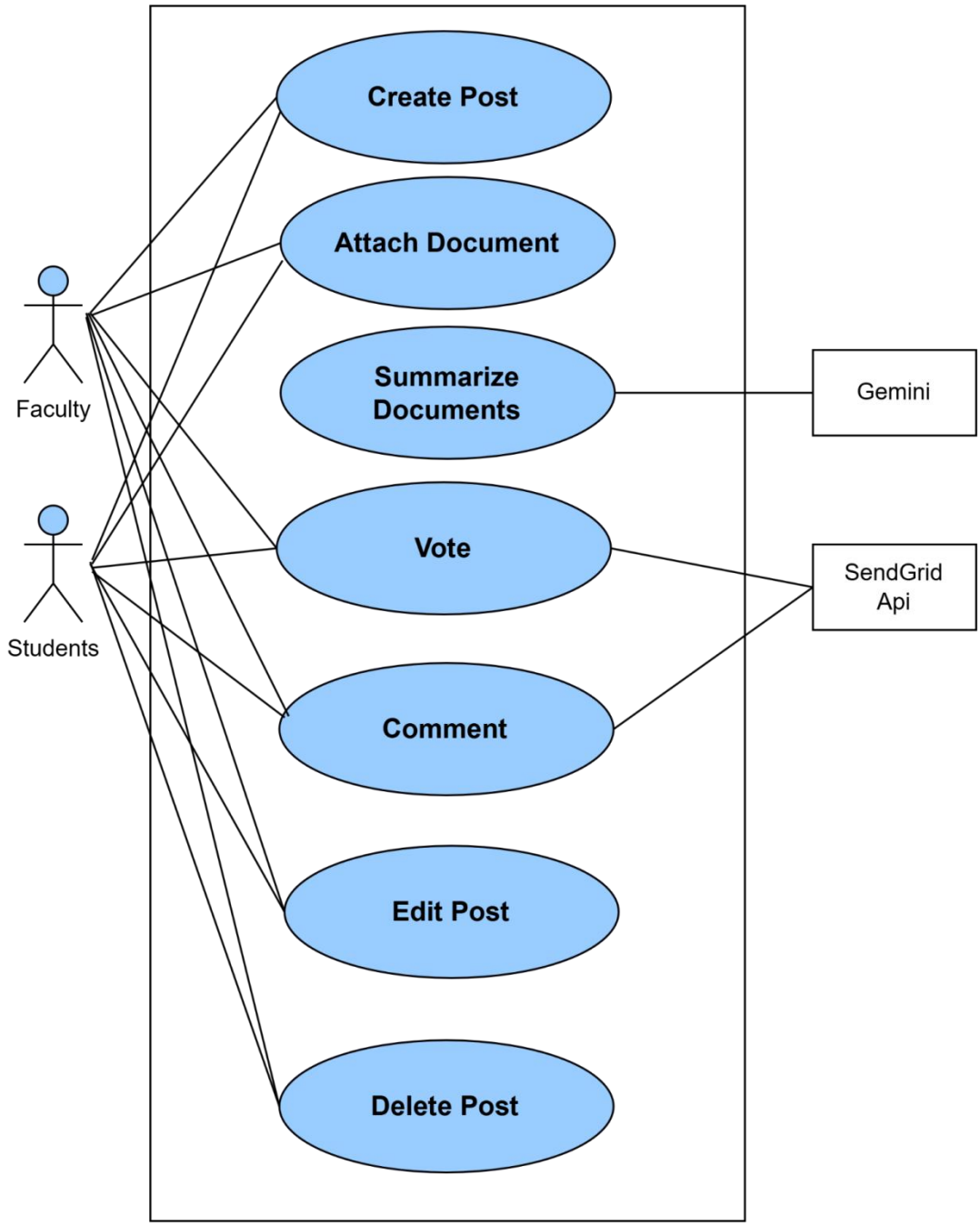


Figure 6. Post use case diagram

Level: 1.5

Name: Message
Primary Actor: Faculty, Students

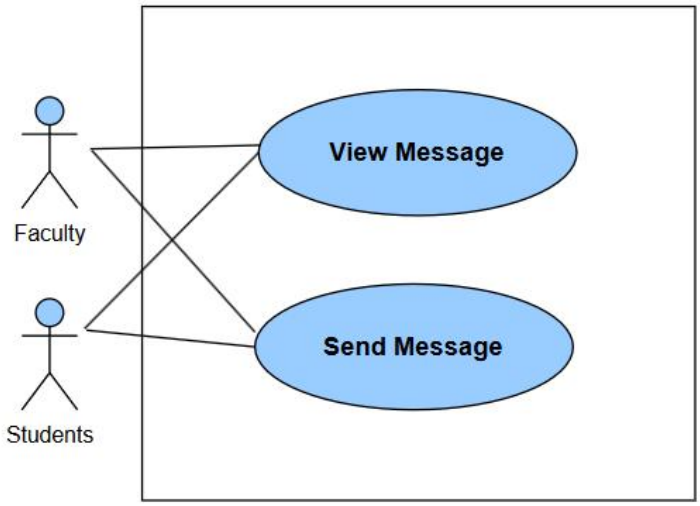


Figure 7. Message use case diagram

Level: 1.6

Name: Gamification
Primary Actor: Faculty, Students

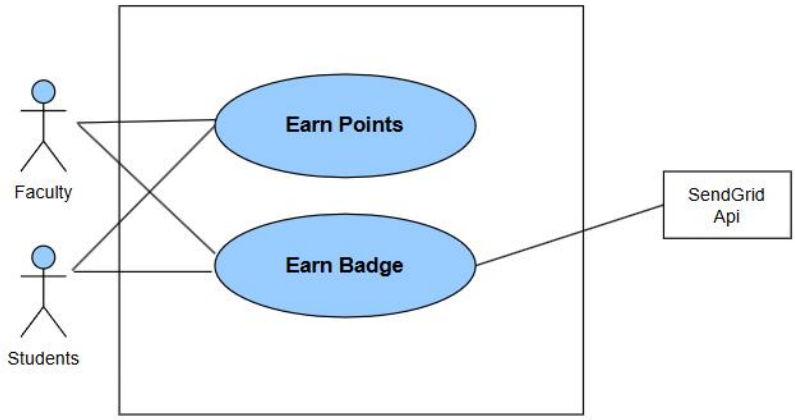


Figure 8. Gamification feed use case diagram

Level: 1.7

Name: Community
Primary Actor: Faculty, Students, Moderator

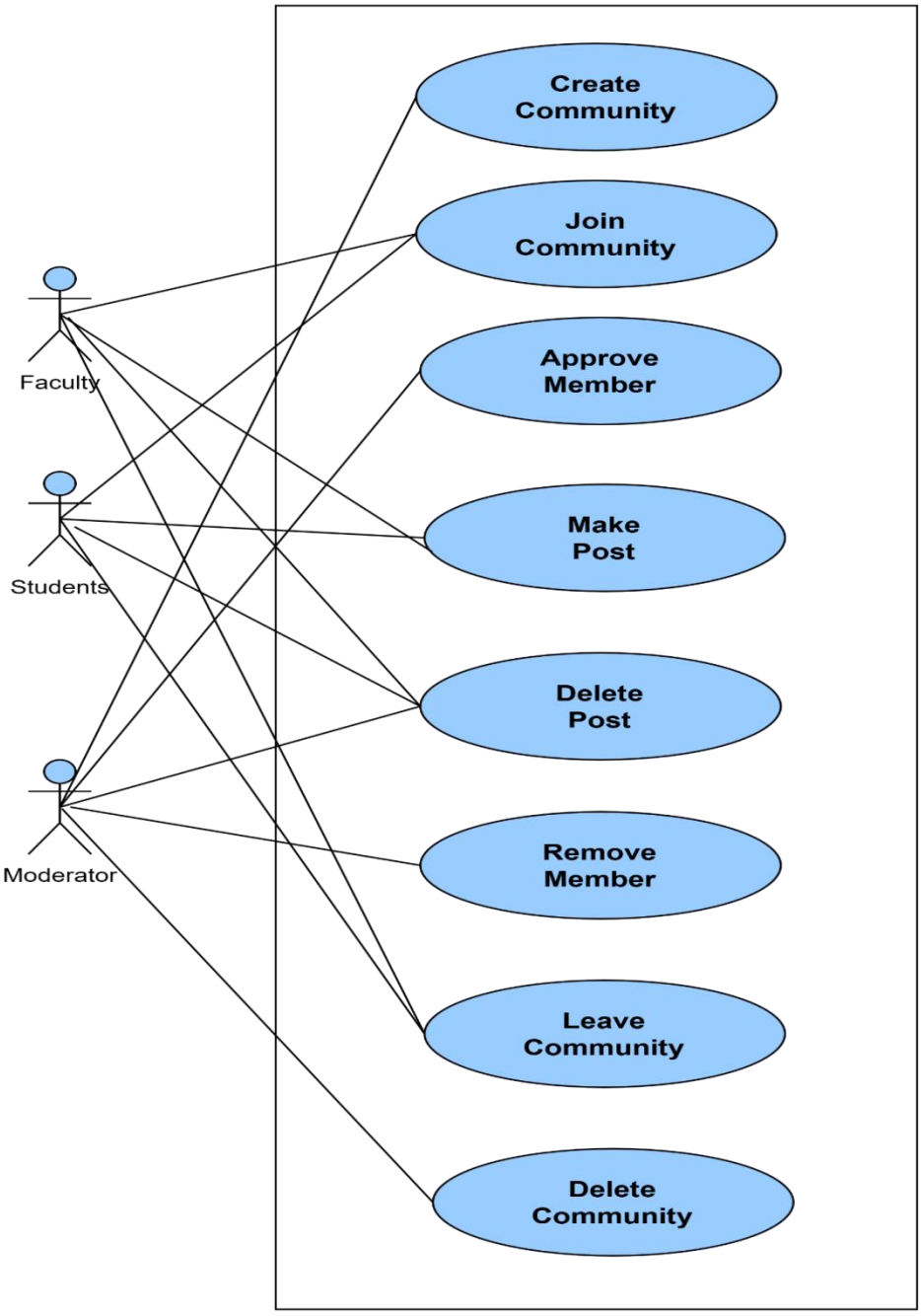


Figure 9. Community use case diagram

Activity Diagram

Level: 1

Name: CampusReConnect(Detailed)
Reference: Use case diagram level 1

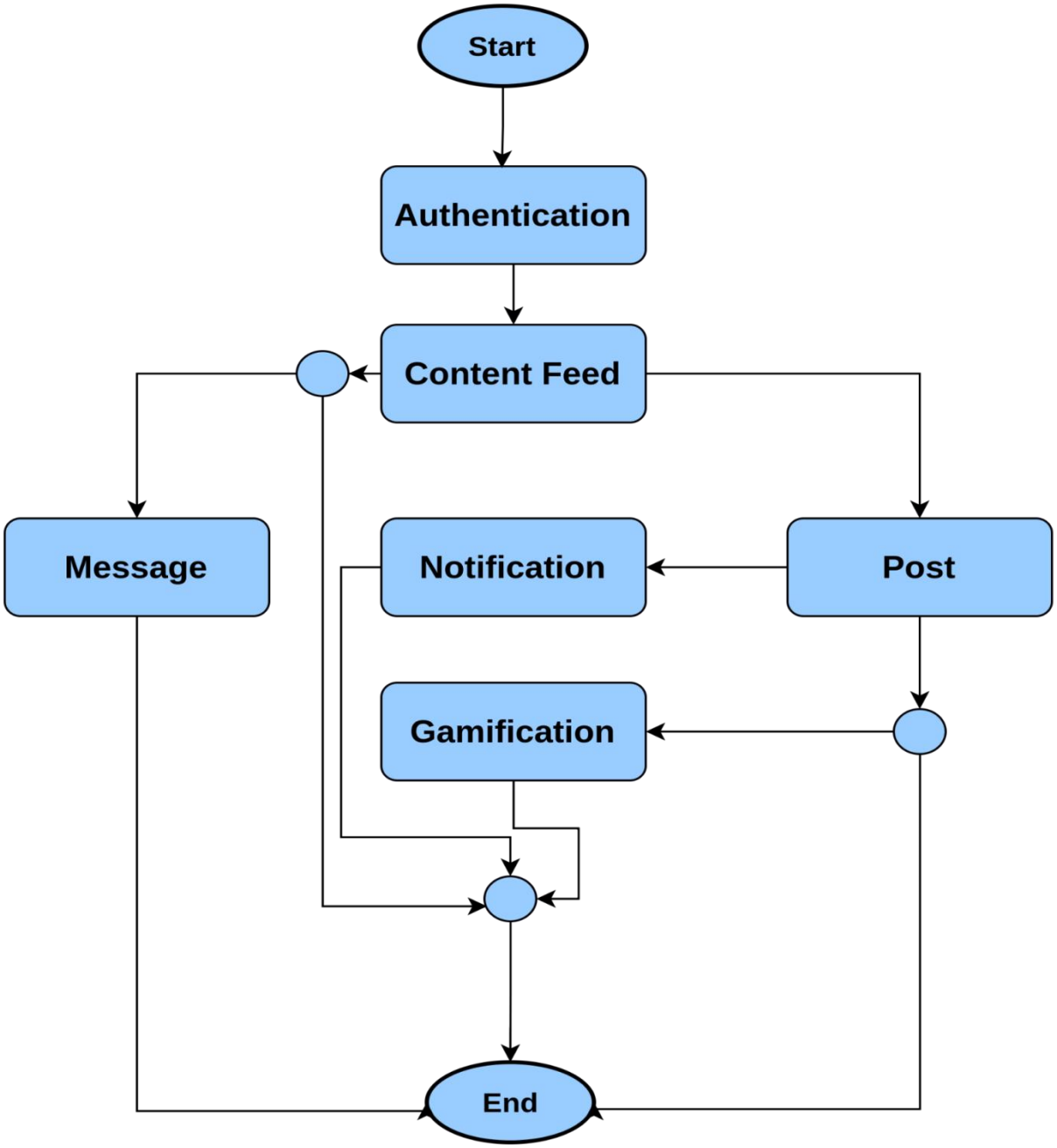


Figure 10. CampusReConnect(Detailed) activity diagram

Level: 1.1

Name: Authentication
Reference: Use case diagram level 1.1

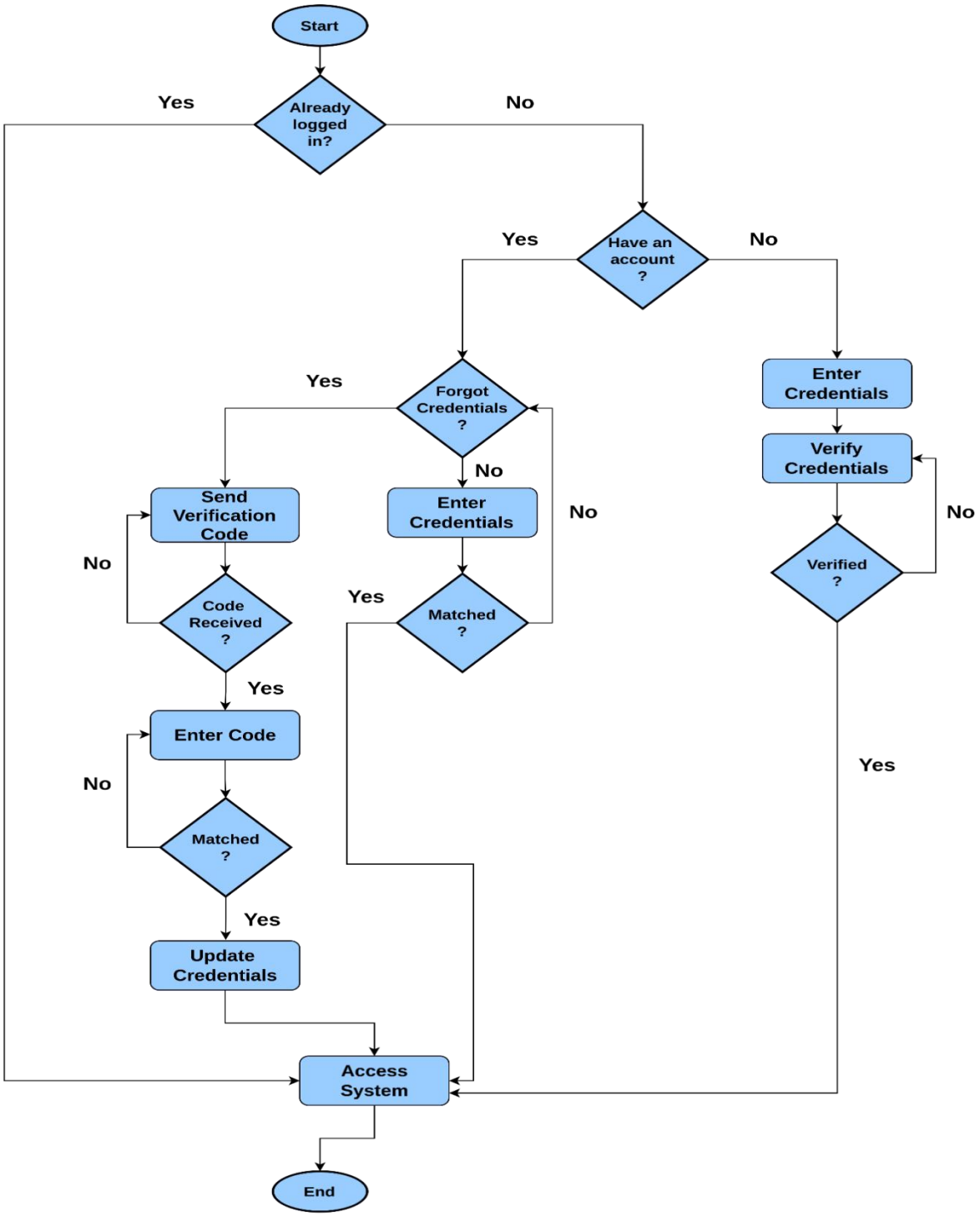


Figure 11. Authentication activity diagram

Level: 1.2
Name: Profile Management
Reference: Use case diagram level 1.2

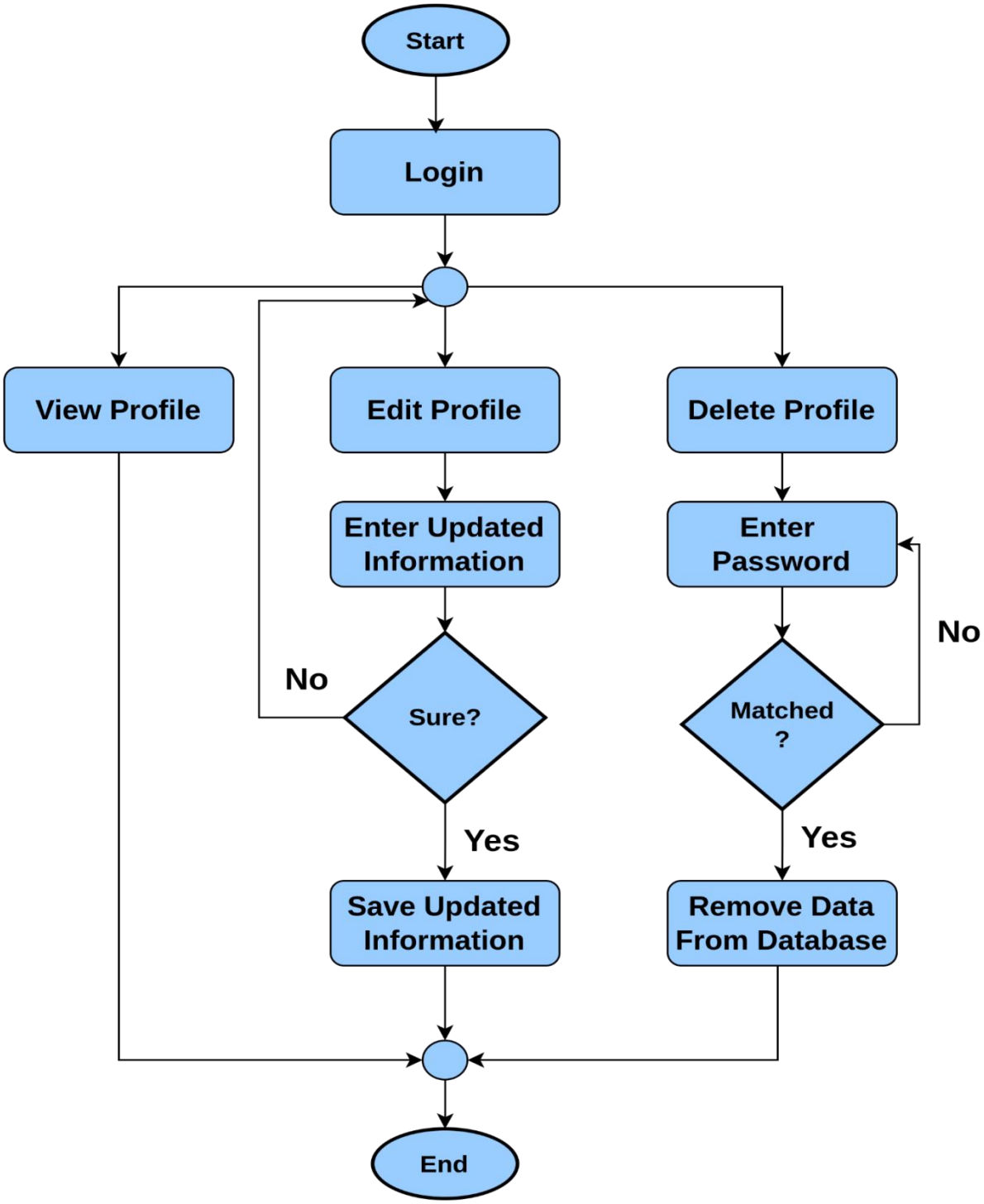


Figure 12. Profile Management activity diagram

Level: 1.3
Name: Content Feed
Reference: Use case diagram level 1.3

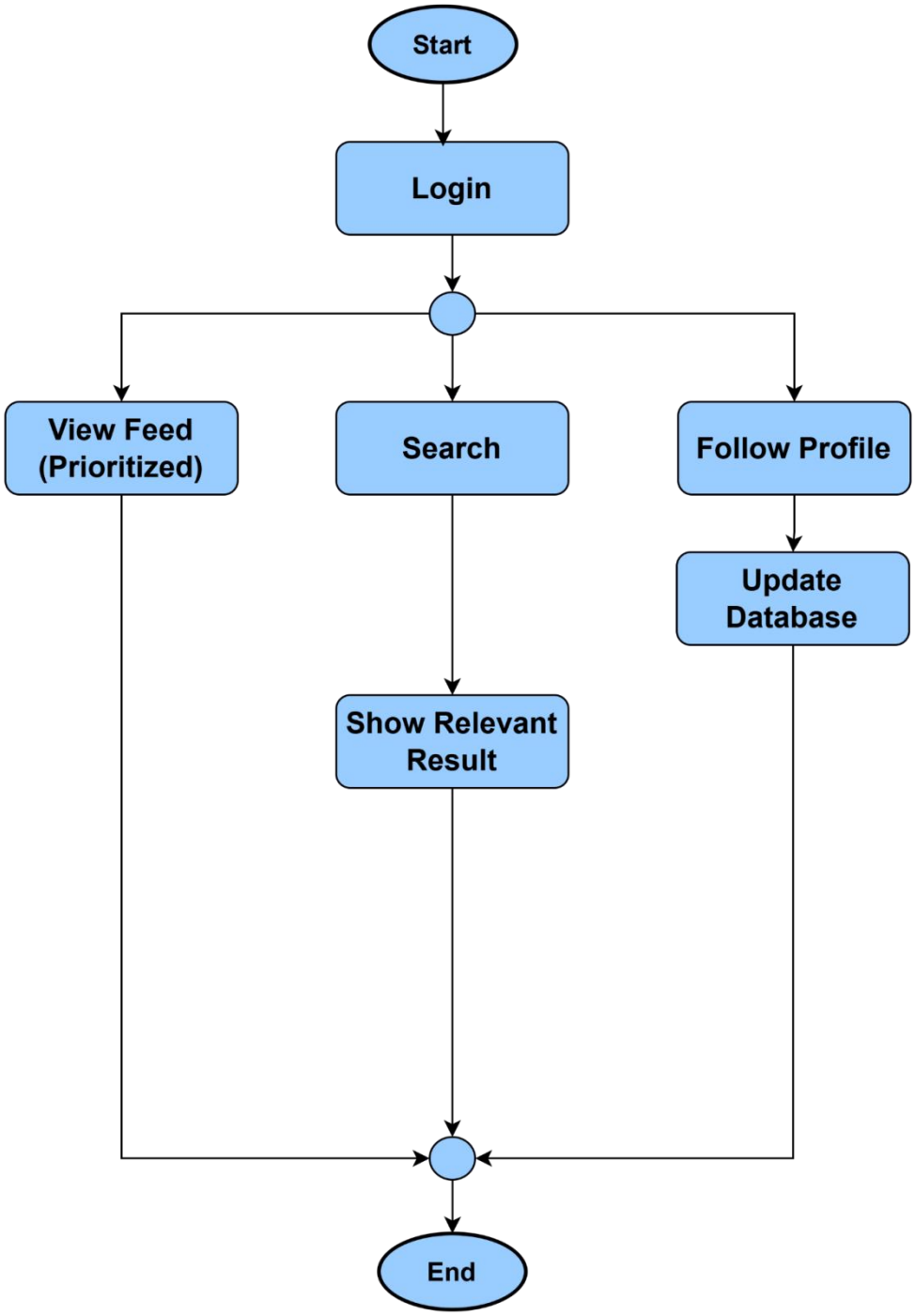


Figure 13. Content Feed activity diagram

Level: 1.4
Name: Post
Reference: Use case diagram level 1.4

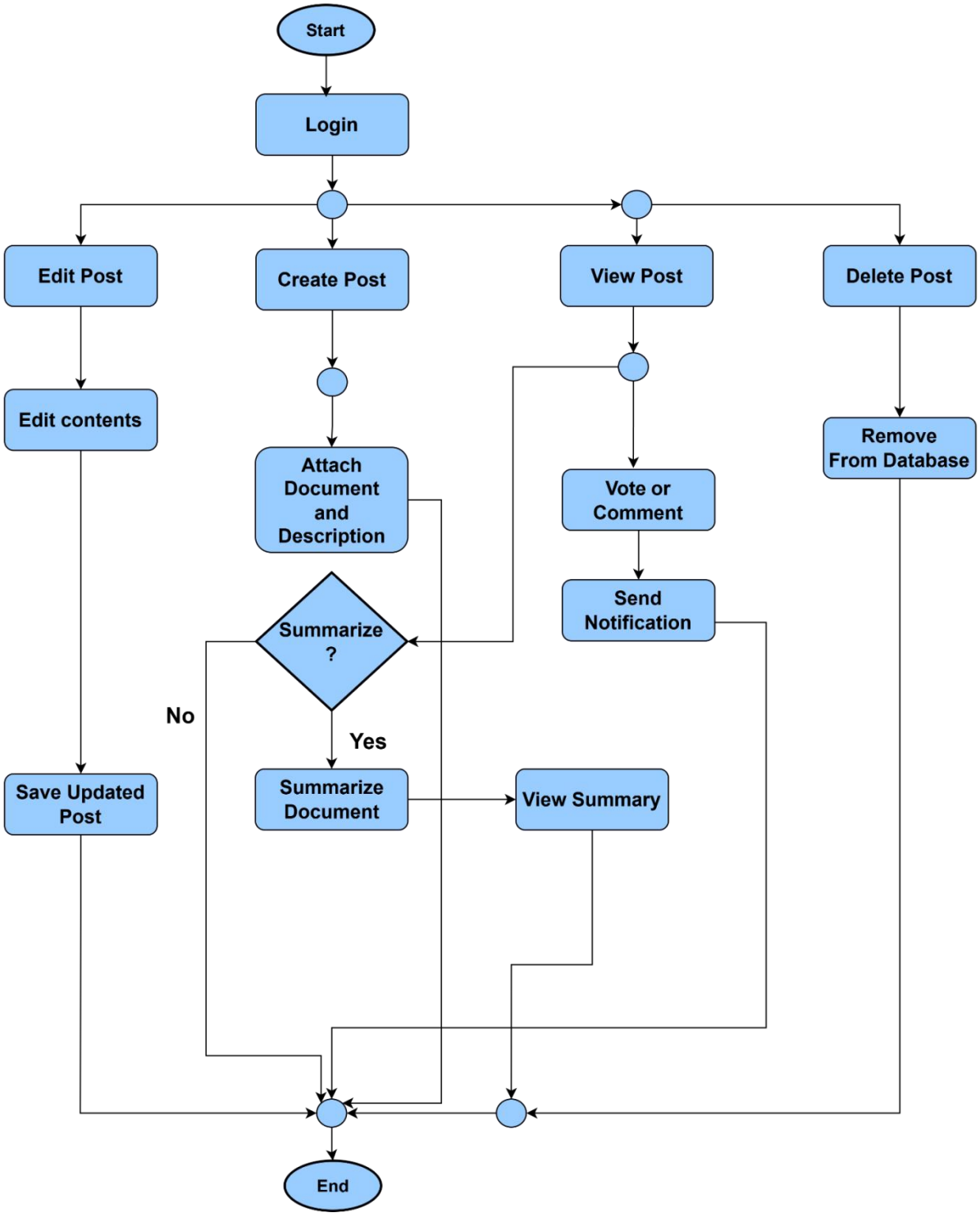


Figure 14. Post activity diagram

Level: 1.5

Name: Message
Reference: Use case diagram level 1.5

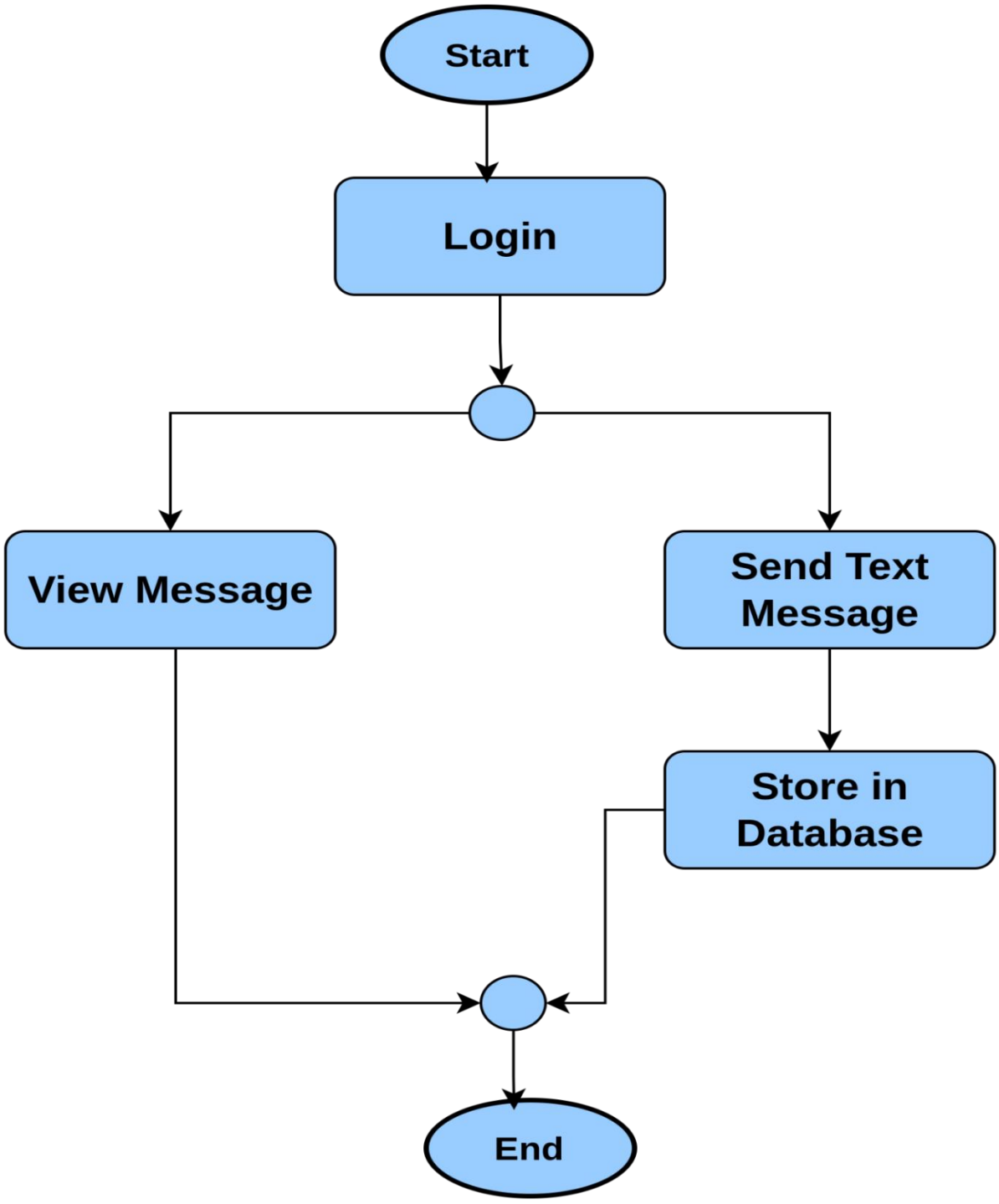


Figure 15. Message activity diagram

Level: 1.6

Name: Gamification
Reference: Use case diagram level 1.6

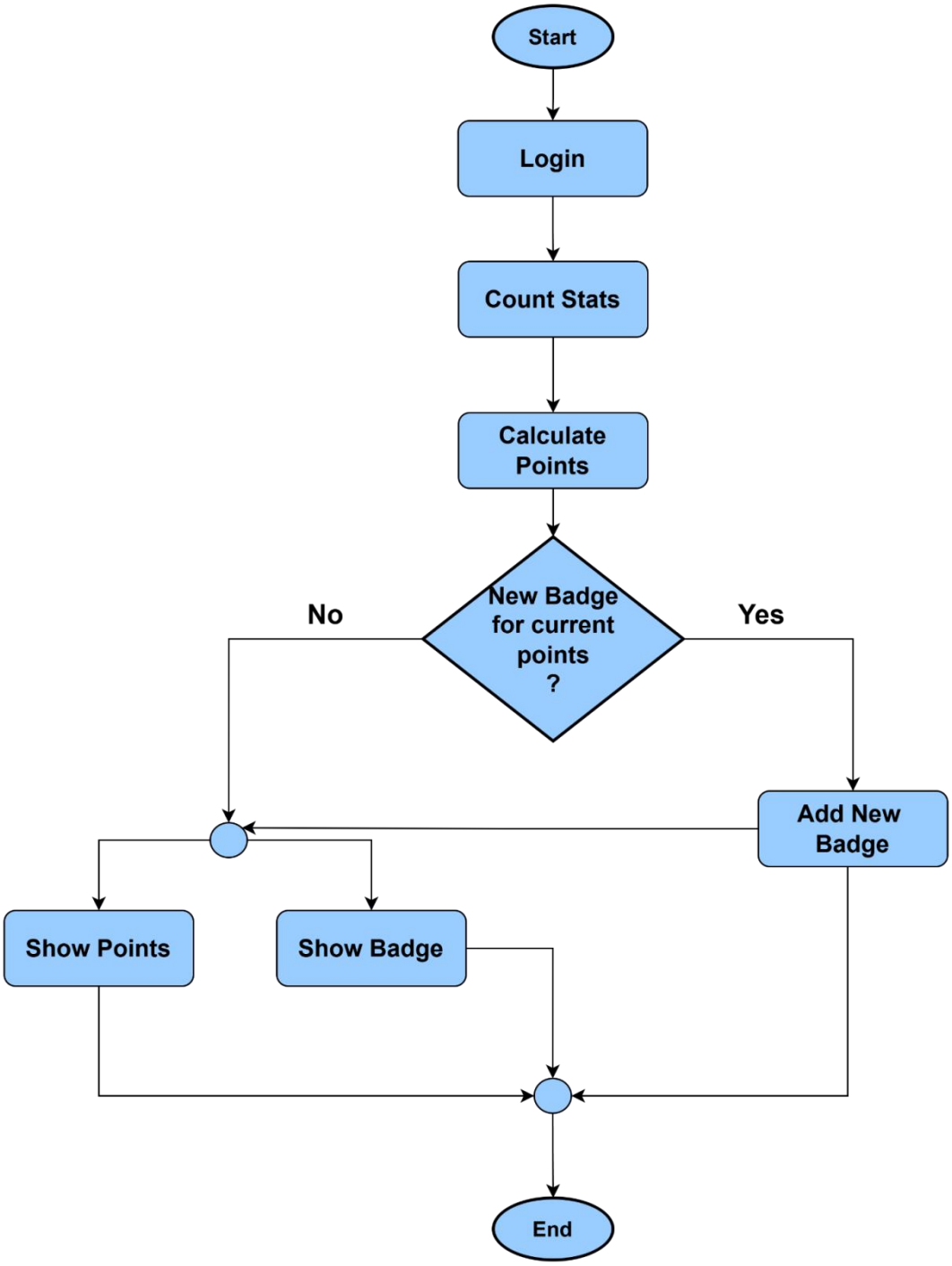


Figure 16. Gamification activity diagram

Level: 1.7

Name: Community
Reference: Use case diagram level 1.7

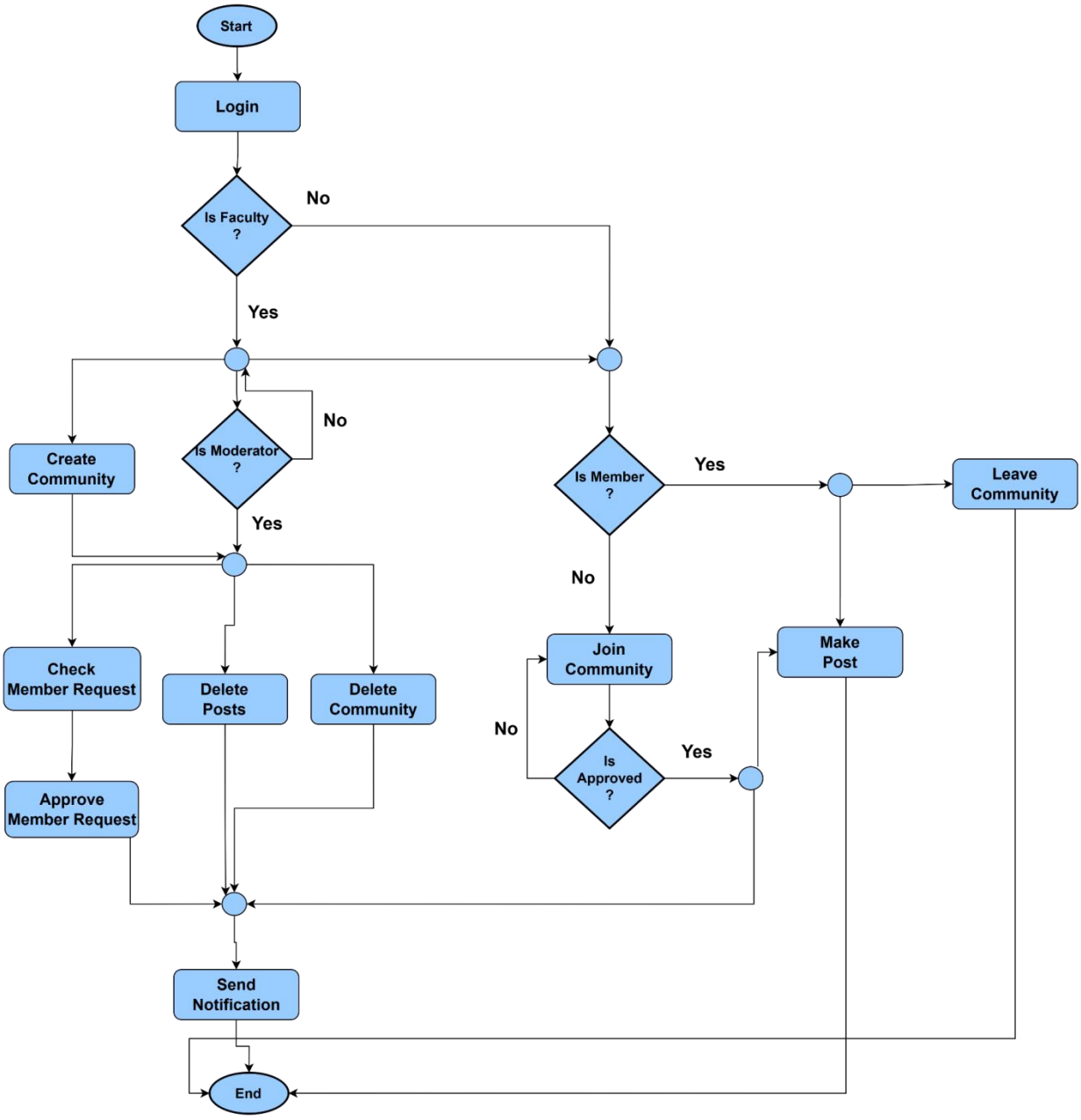


Figure 17. Community activity diagram

Swimlane Diagram

SID (Swimlane ID): 1

Name: CampusReConnect(Detailed)
Reference: Use case and activity diagram level 1

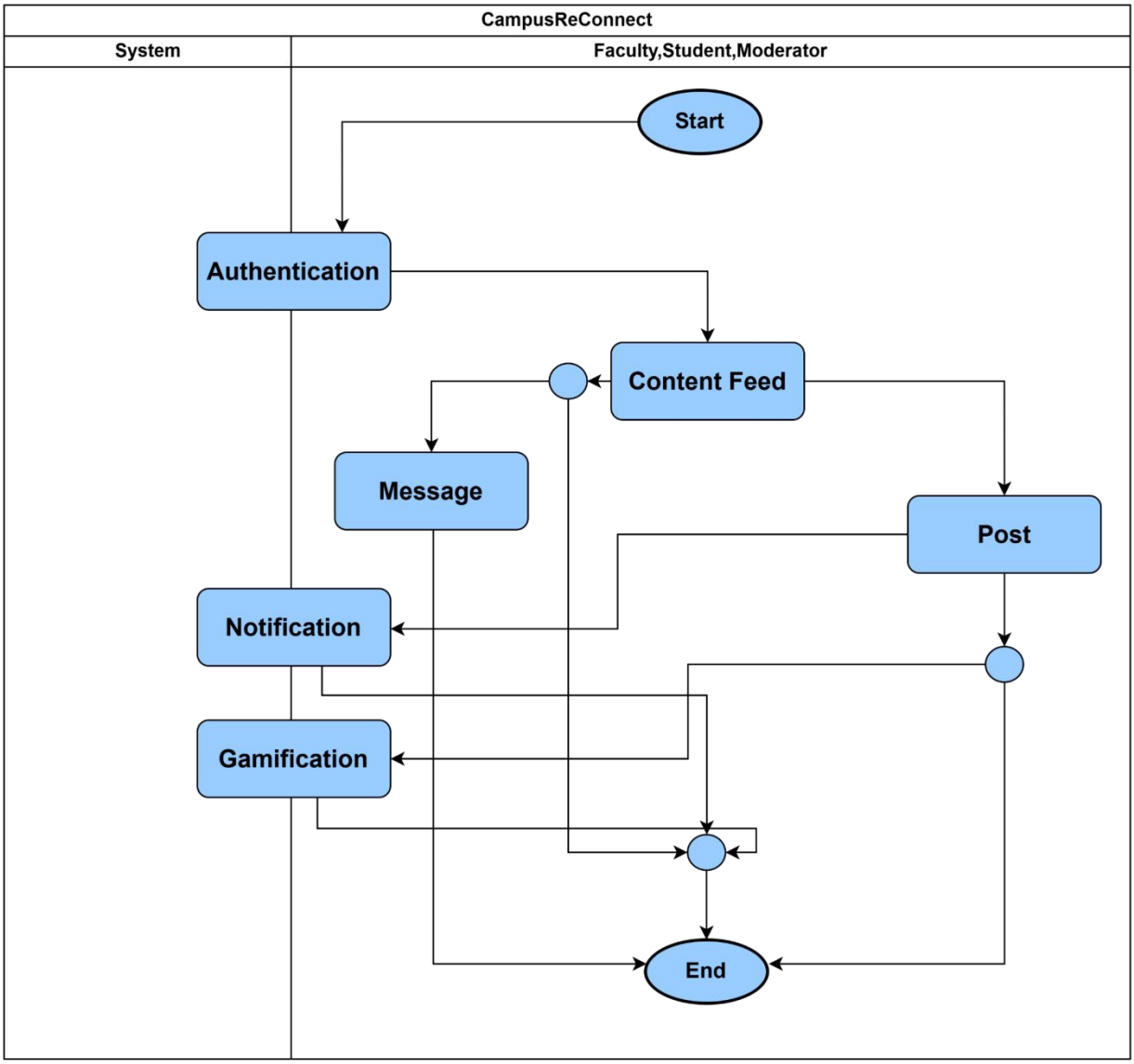


Figure 18. CampusReConnect(Detailed) swimlane diagram

SID (Swimlane ID): 1.1

Name: Authentication
Reference: Use case and activity diagram level 1.1

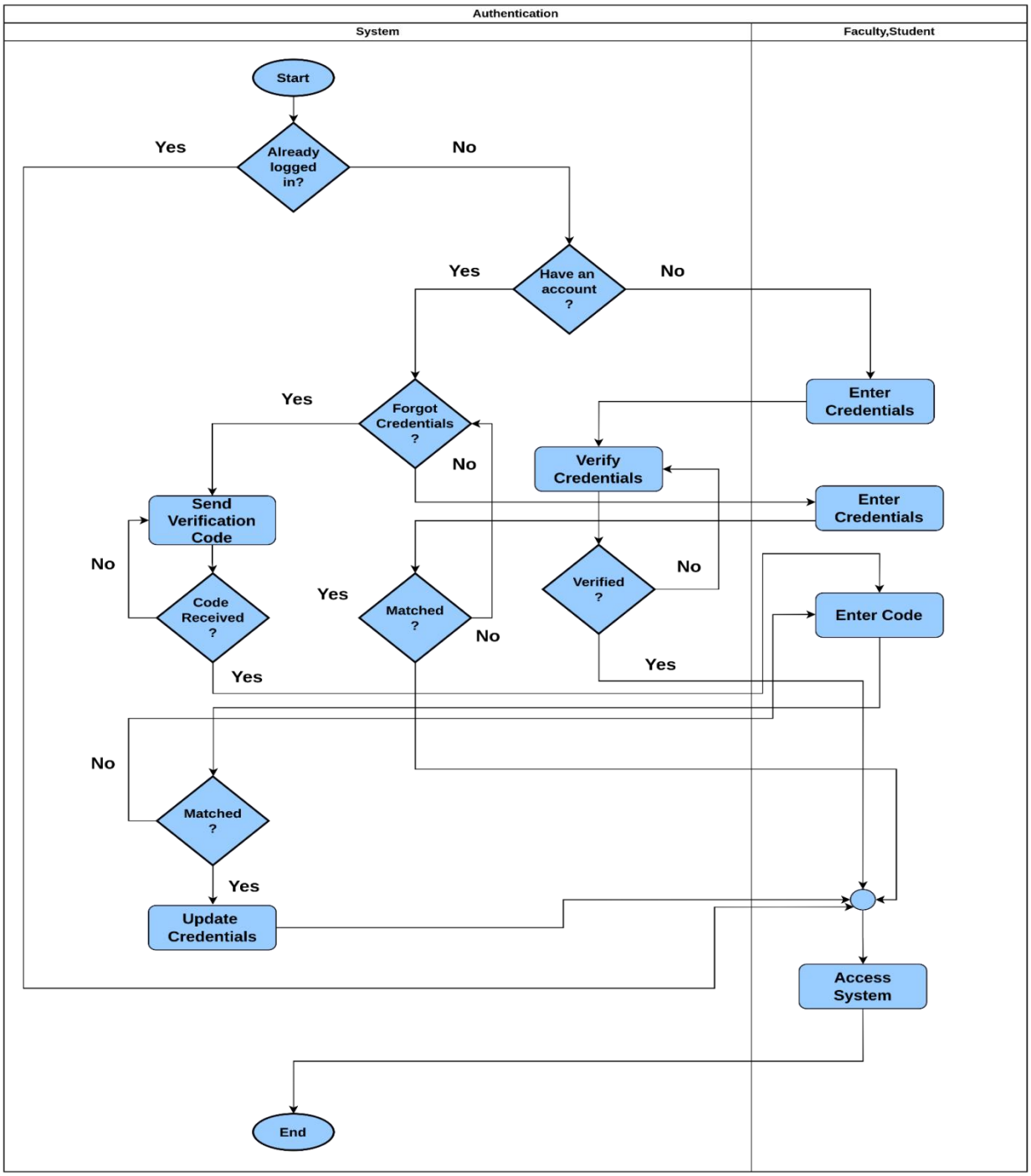


Figure 19. Authentication swimlane diagram

SID (Swimlane ID): 1.2

Name: Profile Management
Reference: Use case and activity diagram level 1.2

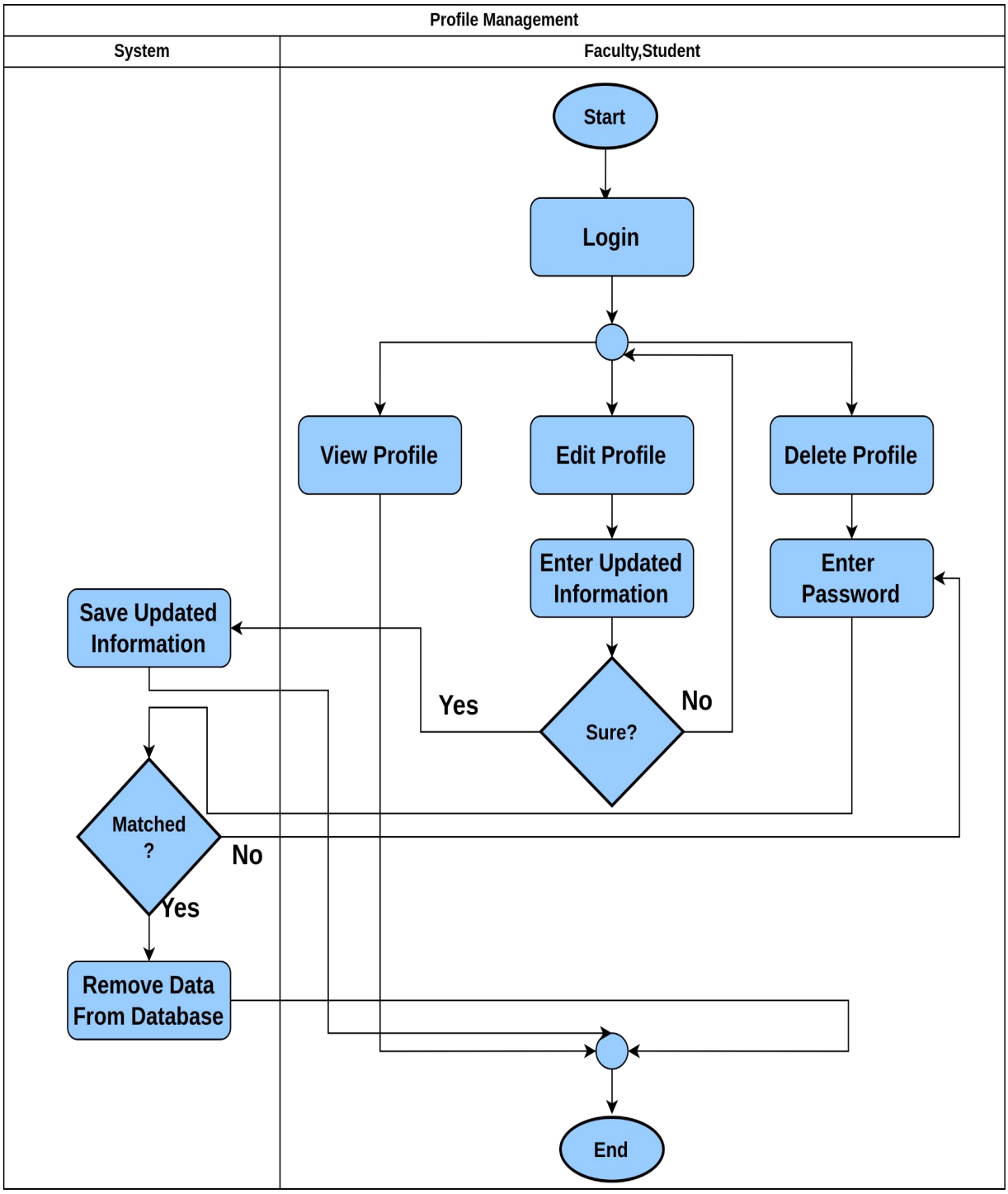


Figure 20. Profile Management swimlane diagram

SID (Swimlane ID): 1.3

Name: Content Feed
Reference: Use case and activity diagram level 1.3

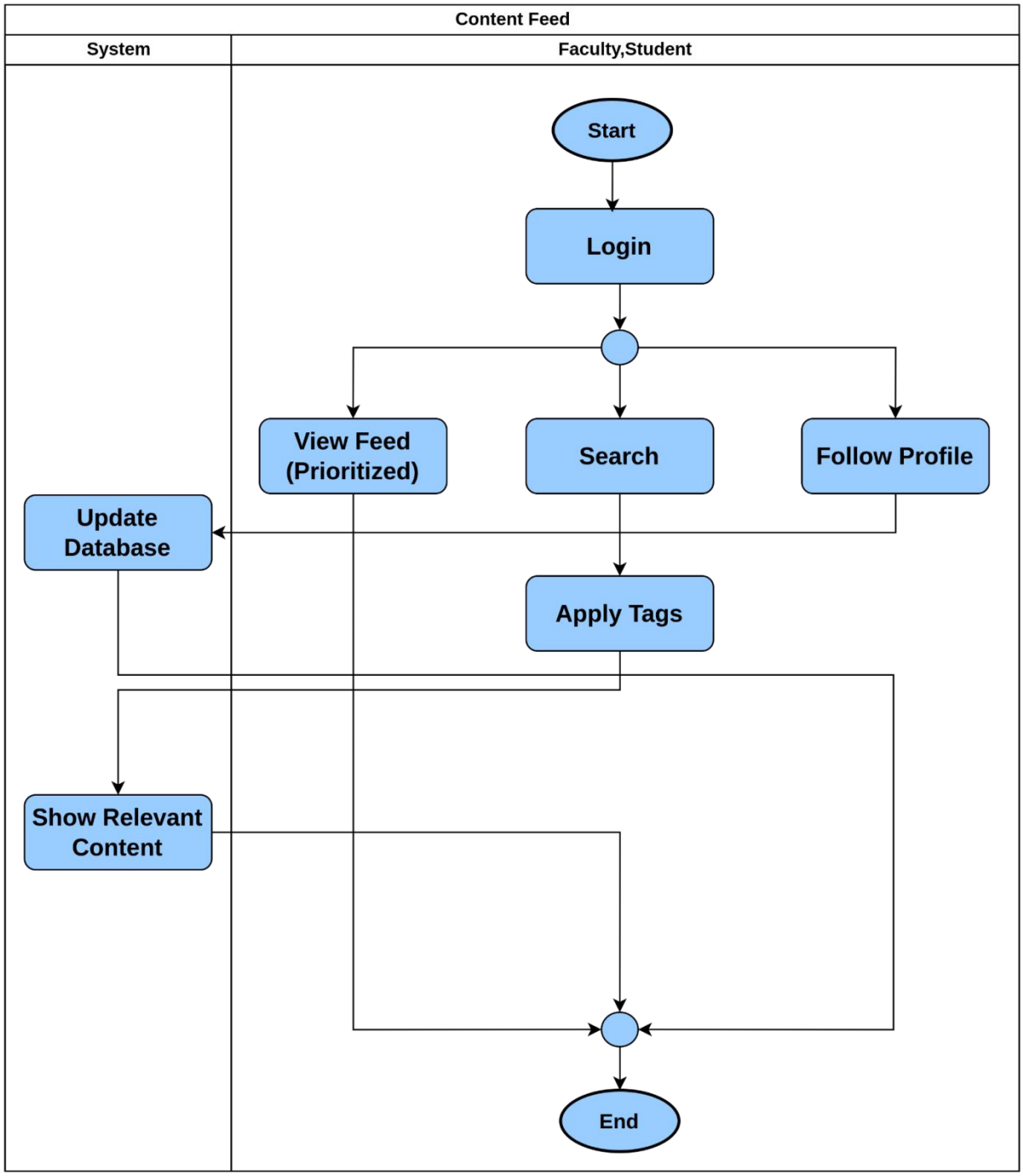


Figure 21. Content Feed swimlane diagram

SID (Swimlane ID): 1.4

Name: Post
Reference: Use case and activity diagram level 1.4

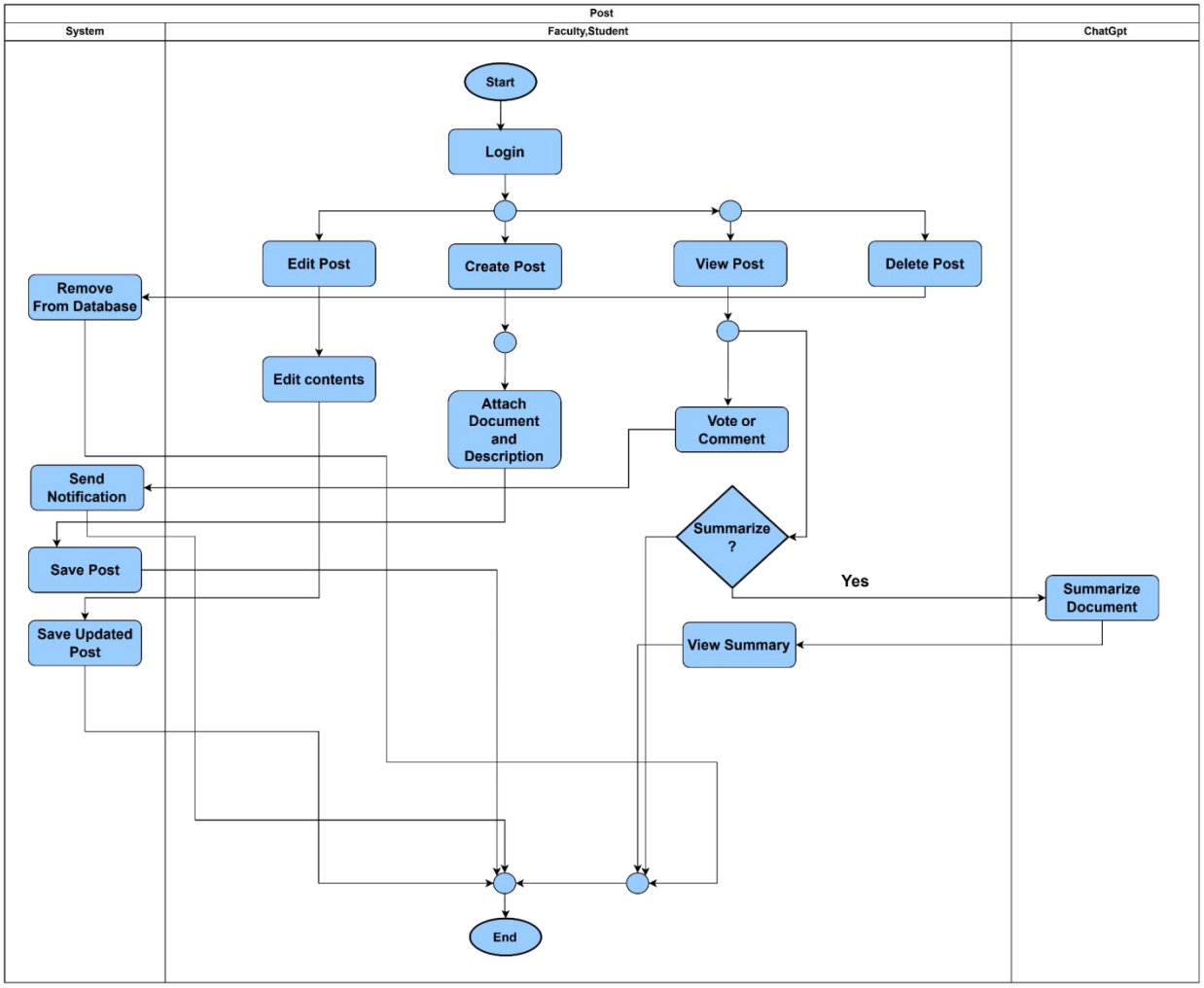


Figure 22. Post swimlane diagram

SID (Swimlane ID): 1.5

Name: Message
Reference: Use case and activity diagram level 1.5

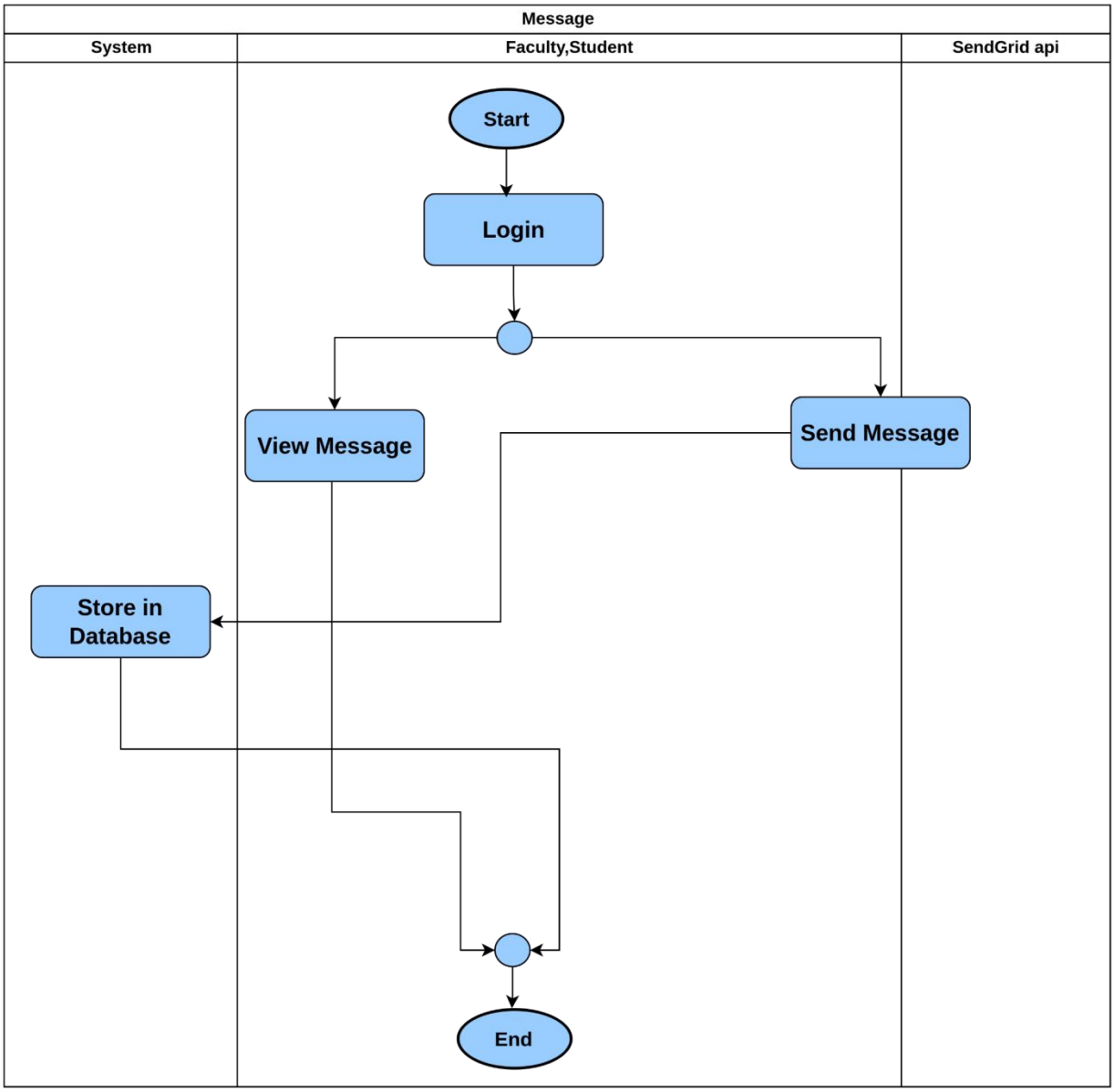


Figure 23. Message swimlane diagram

SID (Swimlane ID): 1.6

Name: Gamification
Reference: Use case and activity diagram level 1.6

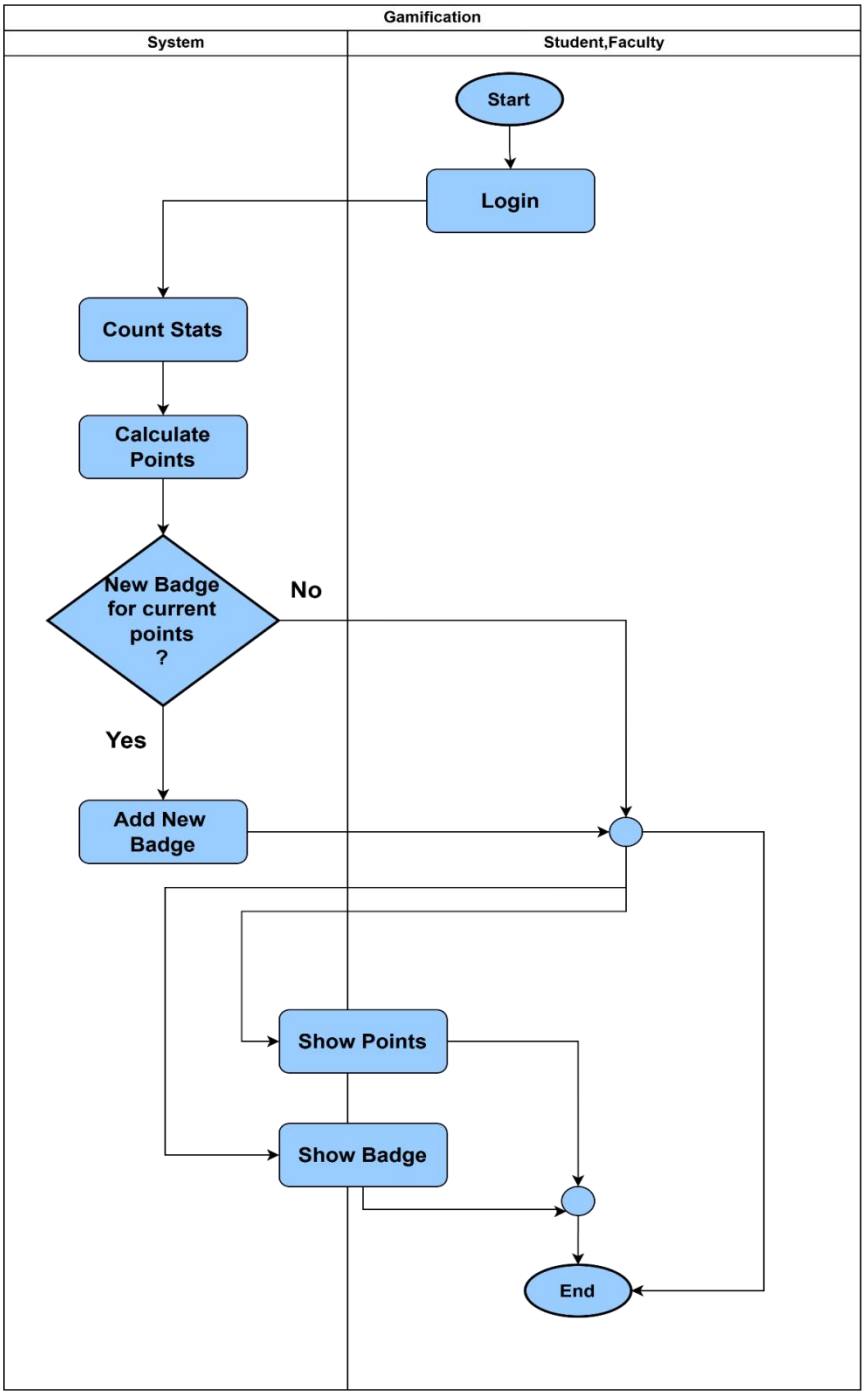


Figure 24. Gamification swimlane diagram

SID (Swimlane ID): 1.7

Name: Community
Reference: Use case and activity diagram level 1.7

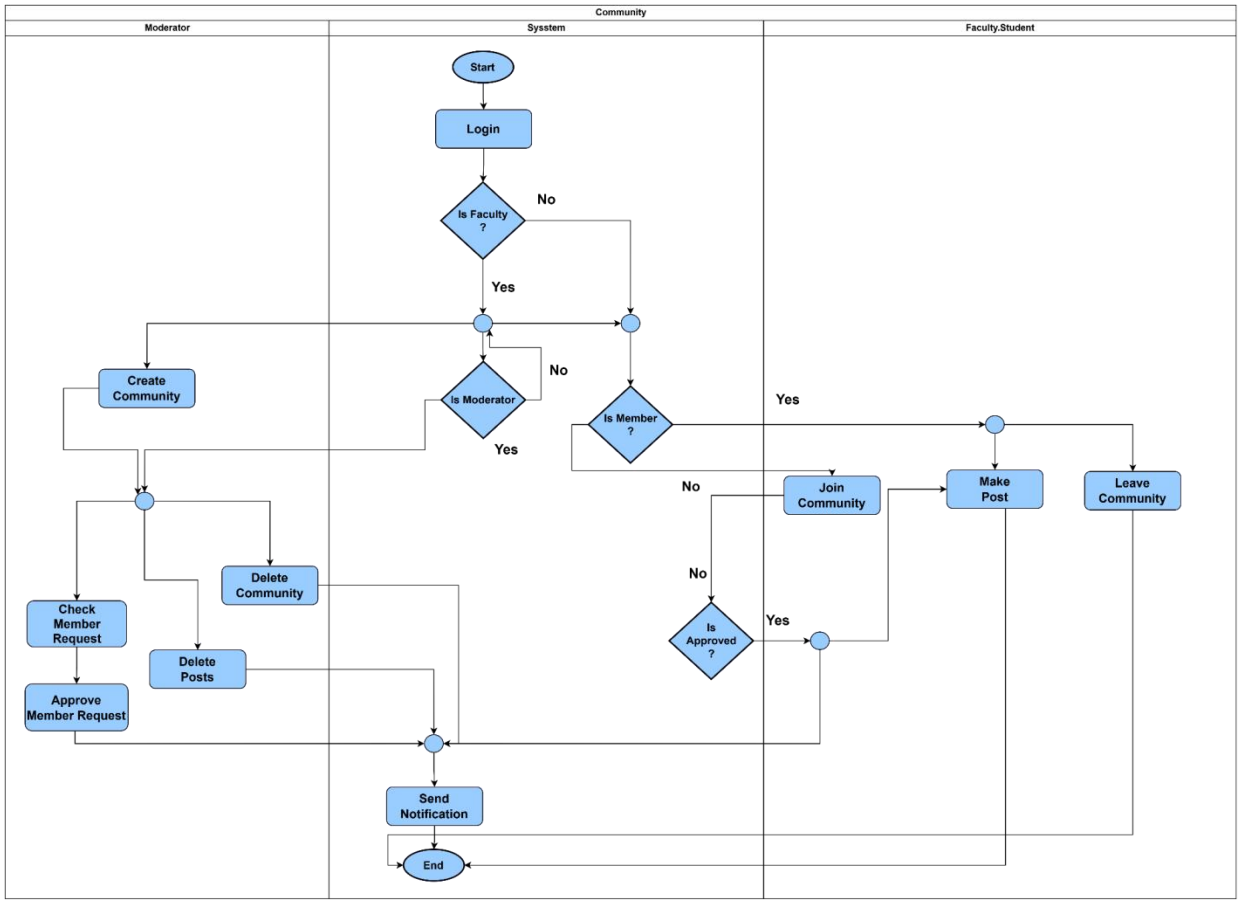


Figure 25. Community swimlane diagram

Data Based Modeling

Data Modeling Concept

A Data Model is an organized view of database concepts and their relationships. The purpose of creating a conceptual data model is to establish entities, their attributes, and relationships. In this data modeling level, there is hardly any detail available on the actual database structure.

The 3 basic elements of Data Modeling are-
- Entity: A real-world thing
- Attribute: Characteristics or properties of an entity
- Relationship: Dependency or association between two entities

The entity relationship diagram (ERD) defines all data objects that are processed within the system, the relationships between the data objects and the information about how the data objects are entered, stored, transformed and produced within the system.

The main goal of a designing data model is to make certain that data objects offered by the functional team are represented accurately. It should be detailed enough to be used for building the physical database.

Data Objects

A data object is a collection of one or more data points that create meaning as a whole. In other words, it's a storage or container to store data values. More specifically, data objects are usable, functional, and meaningful artifacts whose form and function is to encode data. It can be used in type checking operations. A data object can be an external entity, a thing, an occurrence, a role, an organizational unit, a place or a structure.

Data Object Identification

SL No.	Noun	P/S	Attributes
1	Student	S	10, 11, 13, 27, 28, 34, 35, 67, 68, 69, 76
2	Faculty Member	S	10, 11, 13, 27, 28, 34, 35, 67, 68, 69, 76
3	University of Dhaka	P	
4	System	S	
5	User	S	10, 11, 13, 27, 28, 34, 35, 67, 68, 76
6	Community	S	45, 75, 77, 83, 86, 91
7	Account	P	13
8	Platform	P	
9	Email	S	
10	Name	S	
11	Department	S	
12	Password	S	
13	Credentials	S	9, 12, 83
14	Database	P	
15	Confirmation Email	P	
16	Identity	P	
17	Registration	P	
18	Verification	P	
19	Authentication	P	
20	Log in	P	
21	Option	P	
22	Feature	P	
23	Security	P	
24	Privacy	P	

25	Profile	P	
26	Background	P	
27	Research Interest	S	
28	Achievements	S	
29	Information	P	
30	Journey	P	
31	Publications	P	
32	Contributions	P	
33	Metrics	P	
34	Citation Count	S	
35	h-index	S	
36	Stream	P	
37	Posts	S	41, 46, 47, 52, 53, 58, 75, 77, 82, 83, 84, 92
38	Discussions	P	
39	Activities	P	
40	Research Topics	P	
41	Tags	S	
42	Areas	P	
43	Moderator	S	
44	Information Overload	P	
45	Insights	P	
46	Title	S	
47	Description	S	
48	Content	P	
49	Documents	P	
50	Papers	S	81, 83, 90
51	Context	P	
52	Summaries	S	

53	Visibility	S	
54	Individuals	P	
55	Comments	S	58, 78, 83, 85
56	Questions	P	
57	Feedback	P	
58	Reactions	S	59, 60
59	Downvotes	S	
60	Upvotes	S	
61	Message	S	64, 79, 83, 88, 89, 91
62	Communication	P	
63	Conversations	P	
64	Message Content	S	
65	Notifications	S	80, 83, 87, 91
66	Replies	P	
67	Points	S	
68	Badges	S	
69	Rewards	S	
70	Expertise	P	
71	Involvement	P	
72	Research	P	
73	Career	P	
74	University	P	
75	Post_id	S	
76	User_id	S	
77	Community_id	S	
78	Comment_id	S	
79	Message_id	S	
80	Notification_id	S	
81	Paper Link	S	

82	Attachment	S	
83	User_id	S	
84	Supervisor	S	
85	Comment_content	S	
86	Community Name	S	
87	Notification Content	S	
88	Sender	S	
89	Receiver	S	
90	Paper Title	S	
91	SendGrid API	S	
92	Gemini API	S	
93	Approval	S	

Final Data Objects

1. Student
2. Faculty Member
3. User
4. Community
5. Post
6. Paper
7. Comments
8. Message
9. Notifications

Relationship Between Data Objects

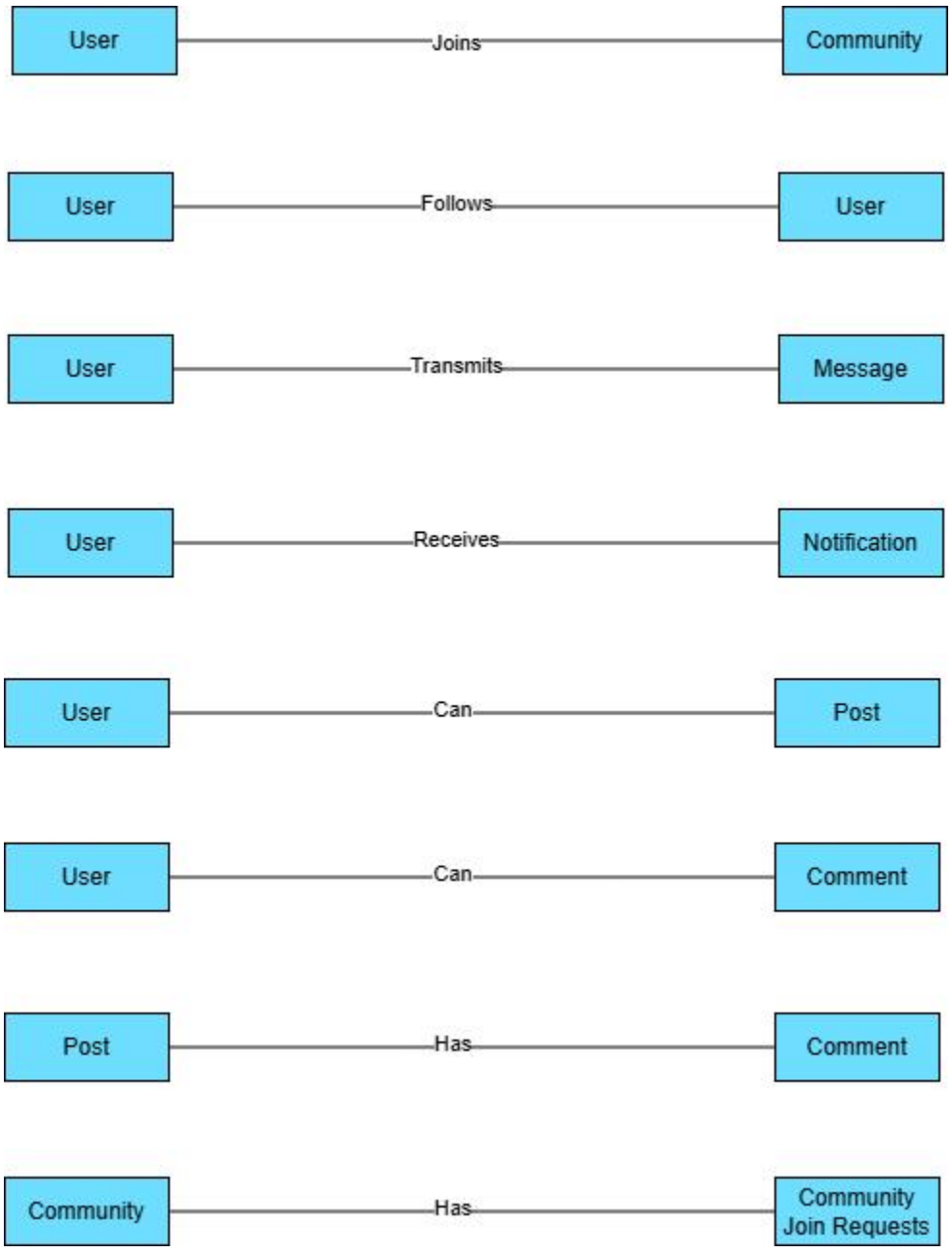


Figure 26. Relationship Between Data Objects

ER Diagram

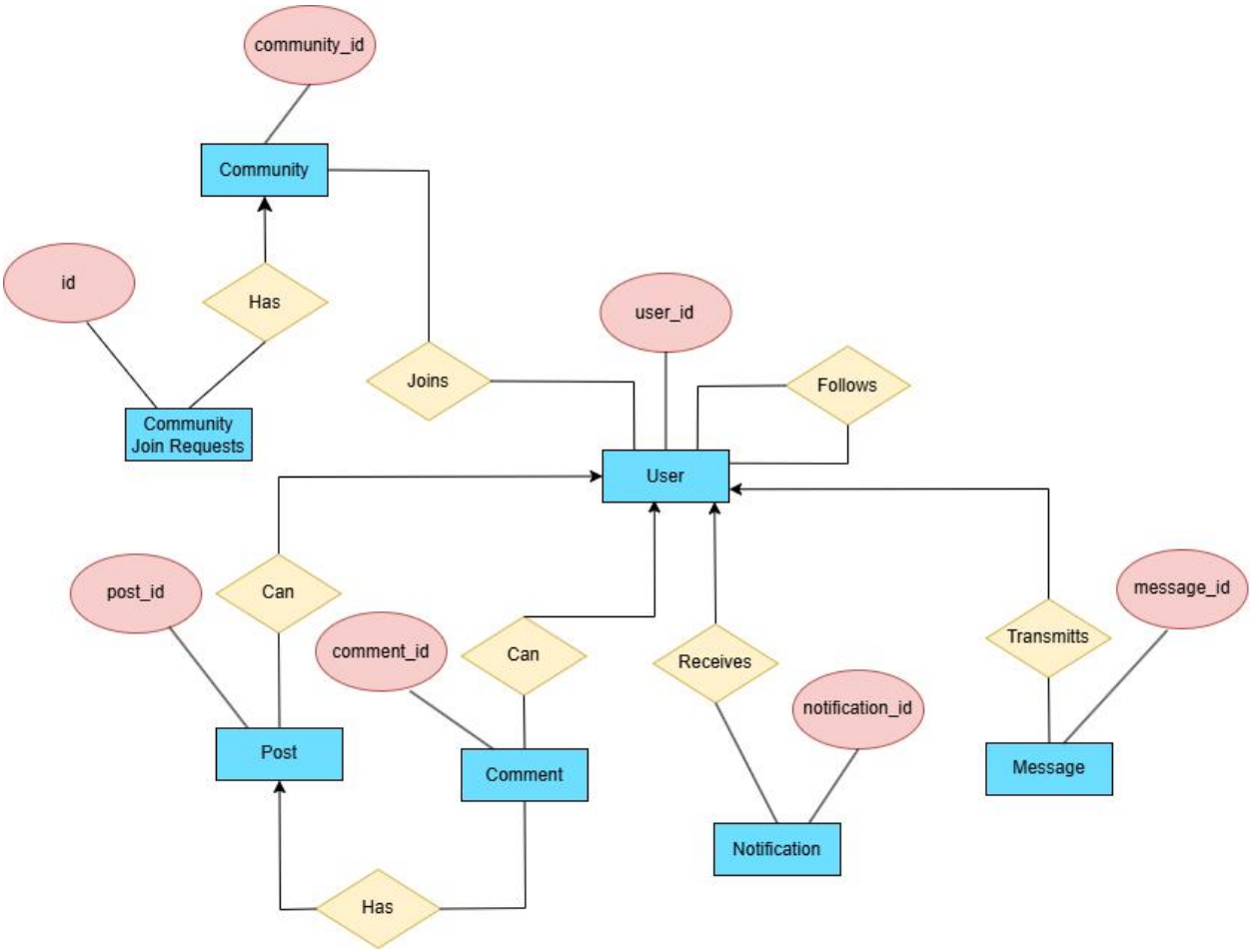


Figure 27. ER diagram

Schema Diagram

Data Object	Attribute	Type	Size
user	user_id is_student email full_name department passwords citation_count h_index points	varchar bit varchar varchar varchar varchar int int varchar	30 1 50 30 30 200 5 5 6

	badge introduction disciplines skillsExpertise languages twitter status	varchar varchar varchar varchar varchar varchar enum	10 1000 1000 1000 1000 100
community	community_id community_name community_description moderator_id	int varchar text varchar	 255 30
post	post_id user_id post_type topic title authors description attachment upvotes downvotes community_id created_date created_time	varchar varchar varchar varchar varchar varchar varchar varchar int int varchar varchar datetime	100 30 30 100 400 100 10000 200 5 5 30 50
comment	comment_id parent_comment id upvotes downvotes user_id post_id comment_content created_date	varchar int int int varchar varchar text varchar	100 5 5 30 100 50
Message	message_id message_content sender_id receiver_id message_time	varchar varchar varchar varchar varchar	100 1000 30 30 100
notification	notification_id sender_id receiver_id notification_content seen	varchar varchar varchar varchar boolean	50 30 30 1000
follow	followee_id follower_id	varchar varchar	30 30

communnity_join_request	id user_id community_id request_time	int varchar int timestamp	30
user_community	id user_id community_id	Int varchar int	30

Class Based Modeling

Class Based Modeling Concept

Class-based modeling identifies classes, attributes and relationships that the system will use. It represents the object. The system manipulates the operations.

The elements of the class based model consist of classes and object, attributes, operations, class – responsibility - collaborator (CRC) models.

Classes are determined using underlining each noun or noun clause and entering it into the simple table. Attributes are the set of data objects that are defining a complete class within the context of the problem. The operations define the behavior of an object.

Noun List

SL No.	Noun	SL No.	Noun
1	Student	48	Content
2	Faculty Member	49	Documents
3	University of Dhaka	50	Papers
4	System	51	Context
5	User	52	Summaries
6	Community	53	Visibility
7	Account	54	Individuals
8	Platform	55	Comments
9	Email	56	Questions
10	Name	57	Feedback
11	Department	58	Reactions
12	Password	59	Downvotes
13	Credentials	60	Upvotes
14	Database	61	Message
15	Confirmation Email	62	Communication
16	Identity	63	Conversations
17	Registration	64	Message Content
18	Verification	65	Notifications
19	Authentication	66	Replies
20	Log in	67	Points
21	Option	68	Badges
22	Feature	69	Rewards

23	Security	70	Expertise
24	Privacy	71	Involvement
25	Profile	72	Research
26	Background	73	Career
27	Research Interest	74	University
28	Achievements	75	Post_id
29	Information	76	User_id
30	Journey	77	Community_id
31	Publications	78	Comment_id
32	Contributions	79	Message_id
33	Metrics	80	Notification_id
34	Citation Count	81	Paper Link
35	h-index	82	Attachment
36	Stream	83	User_id
37	Posts	84	Supervisor
38	Discussions	85	Comment_content
39	Activities	86	Community Name
40	Research Topics	87	Notification Content
41	Tags	88	Sender
42	Areas	89	Receiver
43	Moderator	90	Paper Title
44	Information Overload	91	SendGrid API
45	Insights	92	Gemini API
46	Title	93	Approval
47	Description		

General Classification

Candidate classes were then characterized in seven general classes. The seven general characteristics are as follows:

1. External entities
2. Things
3. Events
4. Roles
5. Organizational units
6. Places
7. Structures

Potential nouns to become a class after general classification criteria

SL No.	Noun	General Class
1	Student	4, 5, 7
2	Faculty Member	4, 5, 7
3	System	7
4	User	4, 5, 7
5	Community	5, 6, 7
6	Email	2
7	Name	2
8	Department	5, 7
9	Password	2
10	Credentials	2
11	Research Interest	2, 5
12	Achievements	2
13	Citation Count	2
14	h-index	2
15	Posts	2, 5, 7
16	Tags	2
17	Moderator	4, 5, 7
18	Title	2
19	Description	2
20	Papers	2, 5
21	Summaries	2
22	Visibility	2
23	Comments	3
24	Reactions	3

25	Downvotes	3
26	Upvotes	3
27	Message	2, 3, 7
28	Message Content	2
29	Notifications	2, 3, 7
30	Points	2
31	Badges	2
32	Post_id	7
33	User_id	7
34	Community_id	7
35	Comment_id	7
36	Message_id	7
37	Notification_id	7
38	Paper Link	2
39	Attachment	2
40	User_id	2
41	Supervisor	2
42	Comment_content	5, 7
43	Community Name	3
44	Notification Content	2, 5
45	Sender	2, 5
46	Receiver	2
47	Paper Title	2

48	SendGrid API	1, 2
49	Gemini API	1, 2
50	Database	1, 2
51	Reward	2, 3, 5
52	Approval	3
53	Authentication	3, 5, 7
54	System	3, 5, 7

Selection Criteria

The candidate classes are then selected as classes by six Selection Criteria. A candidate class generally becomes a class when it fulfills around three characteristics.

1. Retain information
2. Needed services
3. Multiple attributes
4. Common attributes
5. Common operations
6. Essential requirements

Potential general classified nouns to become a class after selection criteria

SL No.	Noun	Selection Criteria
1	Authentication	1-5
2	User	1-5
3	Community	1-5
4	Post	1-5
5	Message	1-5
6	Notification	1-5
7	SendGrid API	1-5
8	Gemini API	1-5
9	Database	1-5
10	Moderator	1-5
11	Gamification	1-5
12	Driver	1-5

Verb List

SL No.	Noun	SL No.	Noun
1	create	36	apply
2	enter	37	enhance
3	send	38	reduce
4	verify	39	share
5	store	40	engage
6	log	41	categorize
7	provide	42	attach
8	authenticate	43	provide
9	choose	44	mention
10	reset	45	leverage
11	forget	46	generate
12	change	47	choose
13	log out	48	interact
14	ensure	49	ask
15	delete	50	react
16	manage	51	upvote
17	update	52	downvote
18	build	53	highlight
19	reflect	54	facilitate
20	highlight	55	notify
21	display	56	post

22	assess	57	receive
23	edit	58	reply
24	keep	59	reward
25	progress	60	answer
26	become	61	engage
27	join	62	accumulate
28	need	63	unlock
29	add	64	reflect
30	approve	65	encourage
31	leave	66	contribute
32	view	67	stay
33	prioritize	68	access
34	follow	69	digest
35	refine	70	enable

Classes

SL No.	Noun	Attributes	Methods
1	Authentication	-userId -email -password -database	+SignUp() +Login() +LogOut() +ChangePassword() +ForgotPassword()

2	User	-userId -isStudent -email -name -department -password -citationCount -hIndex -points -badge -joinedCommunities[]	+followUser() +unfollowUser() +updateStats() +updateProfileTab() +updateProfileSettings() +changePassordSettings() +deleteAccountSettings()
3	Community	-communityId -communityName -moderatorId -userId -pendingMembers[]	+joinCommunity() +leaveCommunity()
4	Post	-postId -userId -tag -title -description -attachment -upvotes -downvotes -communityId -comments[]	+createPost() +editPost() +deletePost() +updatePostVote() +updateCommentVote() +makeComment()
5	Message	-messageId -senderId -receiverId -messageContent -userId	+sendMessages() +viewMessages()
6	Notification	-id -userId	+storeNotifications() +getNotifications()
7	SendGrid API	-senderEmail -receiverEmail -content	+sendEmail()
8	Gemini API	-text	+generateSummary()
9	Database	-user -post -message -notification -community	+SignUp() +updateProfileSettings() +createPost() +editPost() +deletePost() +sendMessage()

			+storeNotification() +searchUser() +getNotification() +viewMessage() +createCommunity()
10	Moderator	-moderatorId -communityId	+createCommunity() +deleteCommunity() +approveMember() +deletePost()
11	Gamification	-userId -points -badge	+calculatePoints() +defineBadge()
12	Driver	-user -post -message -comment -notification -authentication	+SignUp() +isEmailExist() +isStudentEmail() +isFacultyEmail() +Login() +LogOut() +ChangePassword() +sendMessage() +createPost() +editPost() +deletePost() +getNotifications() +searchUser() +getUserStatus() +viewUserList() +getProfileSettings() +getFollowers() +getFollowersCount() +getFollowing() +getFollowingCount() +isFollowingUser() +getComment()

CRC Card

Class	Responsibility	Collaborator
Authentication	● Manage user authentication ● Update database with	User, SendGrid API, Database

	new information	
User	● Manage profile and update own information in database ● Comment and react on posts ● Join and discuss in a community ● Send message to other user	Post, Database, Community, Message, Notification
Community	● Create a space for discussion	User, Moderator, Post, Database
Post	● Show contents ● Create interactions between users	User, Community, Database, Notification
Message	● Build communication between two users	User, Database
Notification	● Send notifications	User, Database, SendGrid API
SendGrid API	● Send message ● Send mail and notification	User, Authentication, Notification
Gemini API	● Summarize content	Post
Database	● Store and manage user, post, community information	User, Post, Community
Moderator	● Approve members in a community ● Create and Manage Community ● Delete irrelevant posts	User, Community, Post
Gamification	● Show the authenticity of the users contributions	User, Database
Driver	● Establish connection between the classes ● Search and filter posts	Authentication, Post, Message Notification, Database

Class Card

Authentication	
Attribute	Method
-userId -email -password -database	+SignUp() +Login() +LogOut() +ChangePassword() +ForgotPassword()
Responsibilities	Collaborator
● Manage user authentication ● Update database with new information	User, SendGrid API Database

User	
Attribute	Method
-userId -isStudent -email -name -department -password -citationCount -hIndex -points -badge -joinedCommunities[]	+followUser() +unfollowUser() +updateStats() +updateProfileTab() +updateProfileSettings() +changePassordSettings() +deleteAccountSettings()

Responsibilities	Collaborator
● Manage profile and update own information in database ● Comment and react on posts ● Join and discuss in a community ● Send message to other user	Post, Database, Community, Message, Notification

Community	
Attribute	Method
-communityId -communityName -moderatorId -pendingMembers[]	+joinCommunity() +leaveCommunity()
Responsibilities	Collaborator
● Create a space for discussion	User, Moderator, Post, Database

Post	
Attribute	Method

-postId -userId -tag -title -description -attachment -upvotes -downvotes -communityId -comments[]	+createPost() +editPost() +deletePost() +updatePostVote() +updateCommentVote() +makeComment()
Responsibilities	Collaborator
• Show contents • Create interactions between users	User, Community, Database, Notification

Message	
Attribute	Method
messageId -senderId -receiverId -messageContent -userId	+sendMessages() +viewMessages()
Responsibilities	Collaborator
• Build communication between two users	User, Database

Notification

Attribute	Method
-id -userId	+storeNotification() +getNotification()
Responsibilities	Collaborator
● Send notifications	User, SendGrid API, Database

SendGrid API	
Attribute	Method
-senderEmail -receiverEmail -content	+sendEmail()
Responsibilities	Collaborator
● Send message ● Send mail and notification	User, Authentication, Notification

Gemini API	
Attribute	Method
-text	+generateSummary()
Responsibilities	Collaborator

• Summarize content	Post

Database	
Attribute	Method
-user -post -message -notification -community	+SignUp() +updateProfileSettings() +createPost() +editPost() +deletePost() +sendMessage() +storeNotification() +searchUser() +getNotification() +viewMessage() +createCommunity()
Responsibilities	Collaborator
• Store and manage user, post, community information	User, Post, Community

Moderator	
Attribute	Method
-moderatorId -communityId	+createCommunity() +deleteCommunity() +approveMember() +deletePost()
Responsibilities	Collaborator

• Approve members in a community • Manage Community • Delete irrelevant posts	User, Community, Post

Gamification	
Attribute	Method
-userId -points -badge	+calculatePoints() +defineBadge()
Responsibilities	Collaborator
• Show the authenticity of the users contributions	User, Database

Driver	
Attribute	Method
-user -post -message -comment -notification -authentication	+SignUp() +isEmailExist() +isStudentEmail() +isFacultyEmail() +Login() +LogOut() +ChangePassword() +sendMessage() +createPost() +editPost() +deletePost() +getNotifications() +searchUser()

	+getUserStatus() +viewUserList() +getProfileSettings() +getFollowers() +getFollowersCount() +getFollowing() +getFollowingCount() +isFollowingUser() +getComment()
Responsibilities	Collaborator
• Establish connection between the classes • Search and filter posts	Authentication, Post, Message, Notification, Database

CRC Diagram:

Diagram ID: 1

Name: Authentication

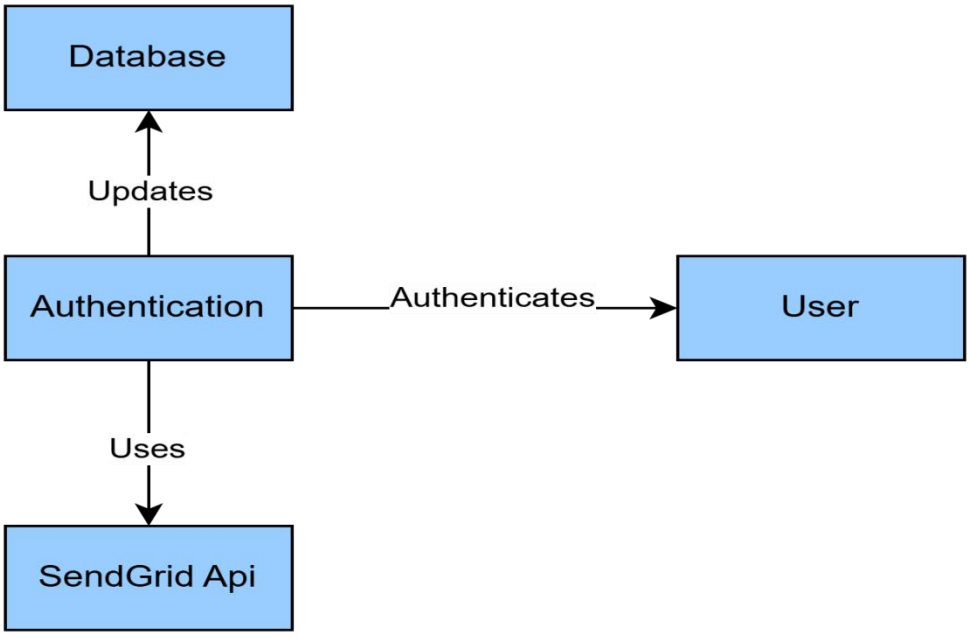


Figure 28. Authentication CRC Diagram

Diagram ID: 2
Name: User

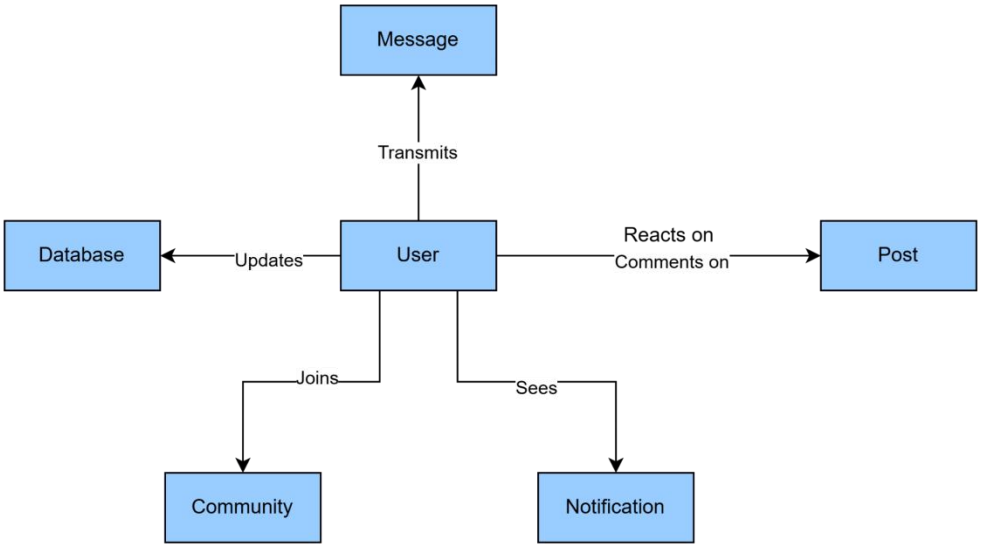


Figure 29. User CRC Diagram

Diagram ID: 3
Name: Community

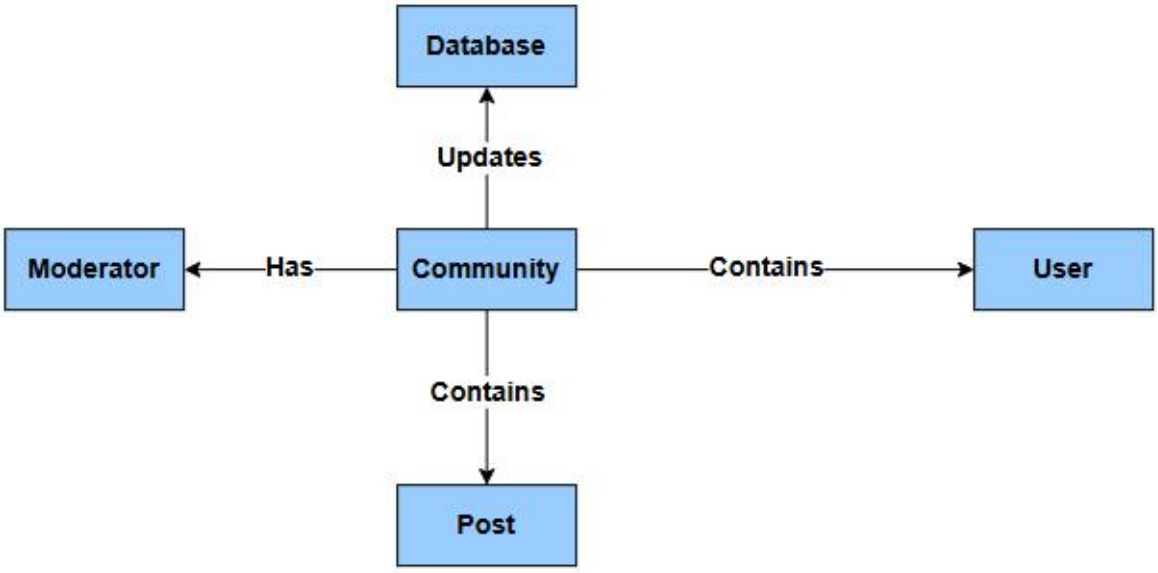


Figure 30. Community CRC Diagram

Diagram ID: 4
Name: Post

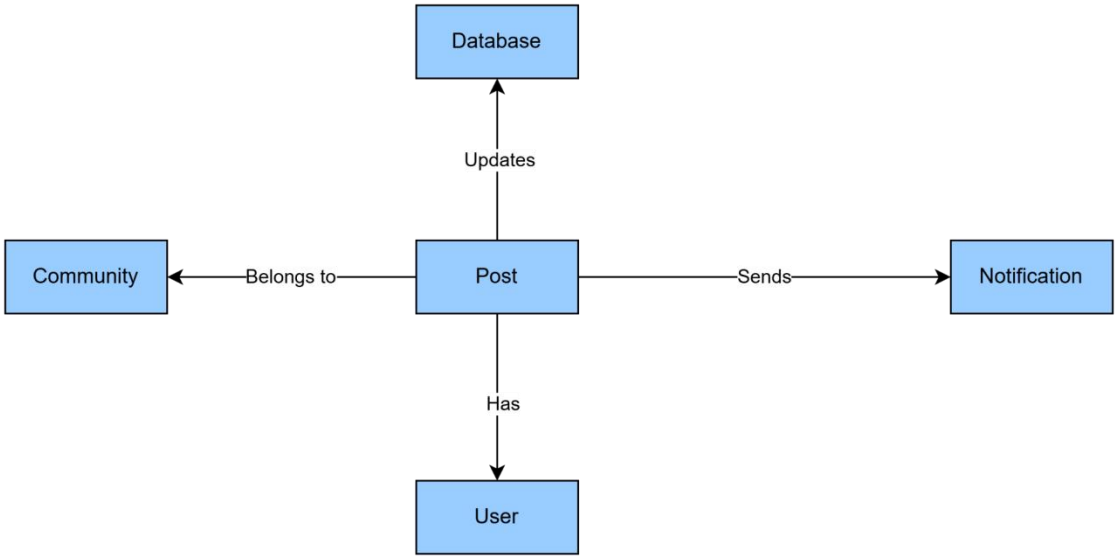


Figure 31. Post CRC Diagram

Diagram ID: 5
Name: Message

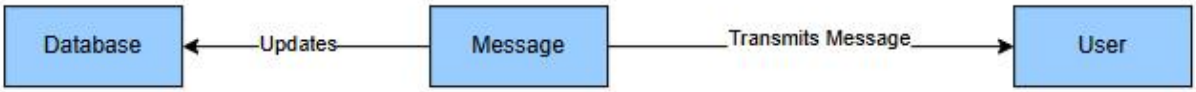


Figure 32. Message CRC Diagram

Diagram ID: 6
Name: Notification

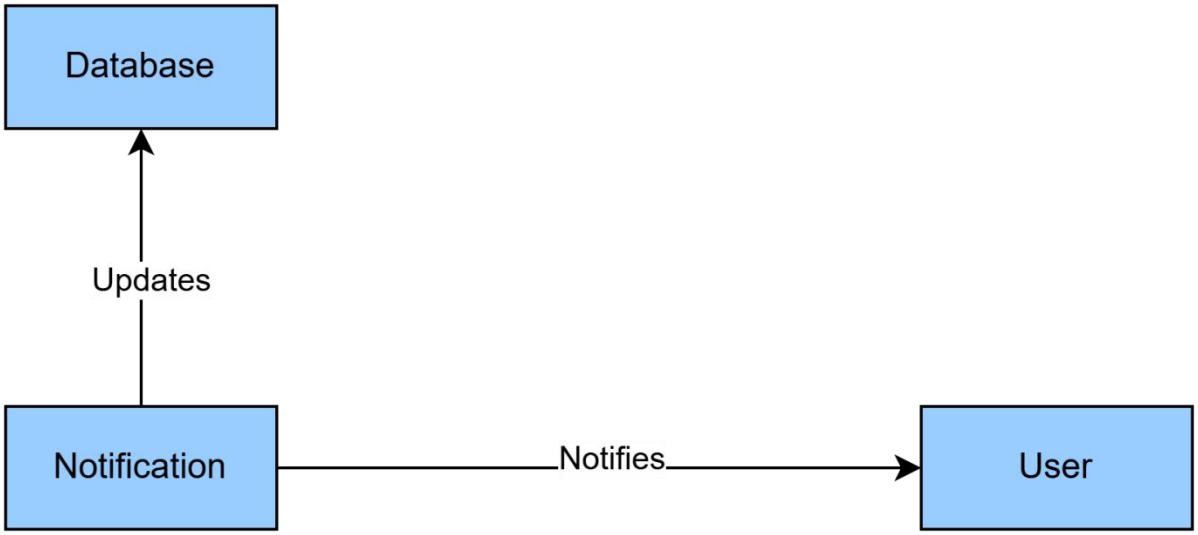


Figure 33. Notification CRC Diagram

Diagram ID: 7
Name: SendGrid API

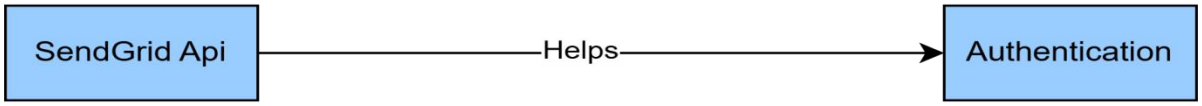


Figure 34. SendGrid API CRC Diagram

Diagram ID: 8
Name: Gemini API

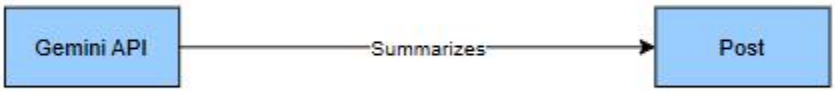


Figure 35. Gemini API CRC Diagram

Diagram ID: 9
Name: Database

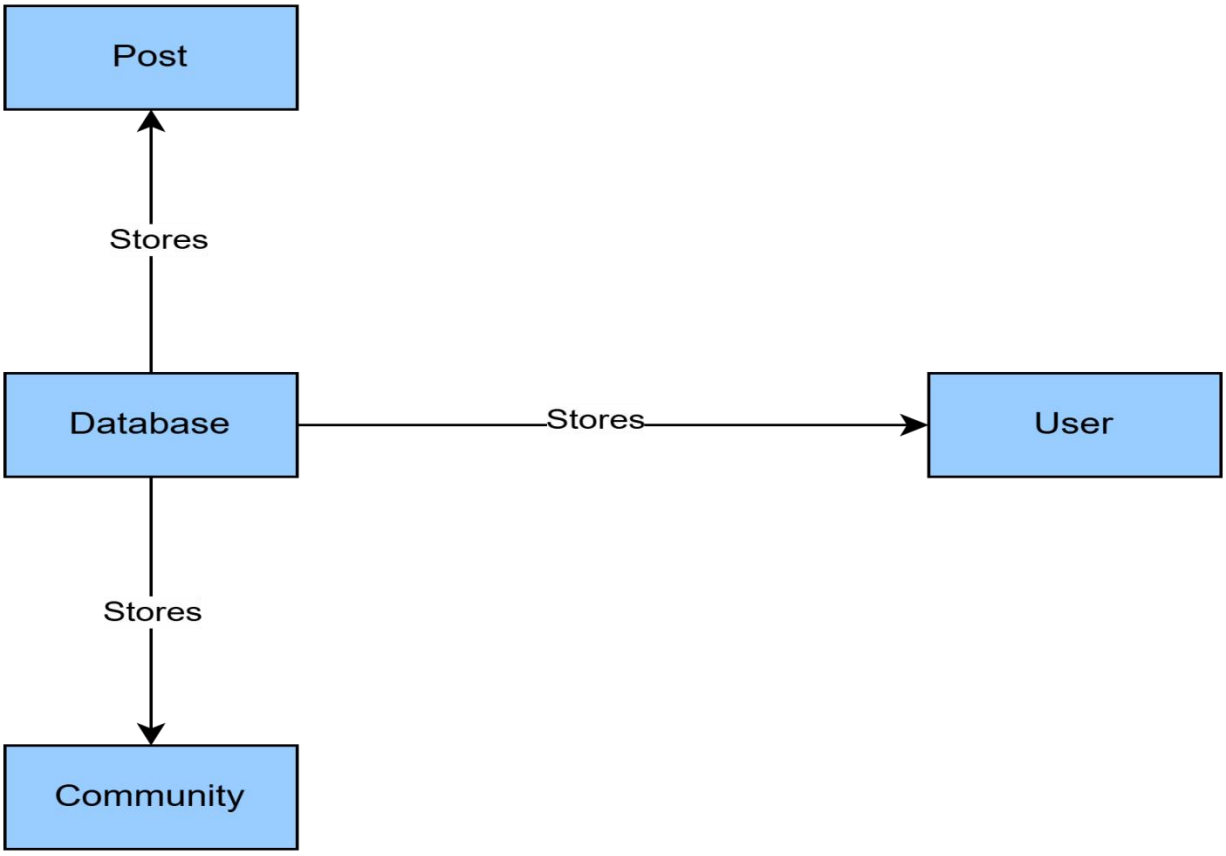


Figure 36. Database CRC Diagram

Diagram ID: 10
Name: Moderator

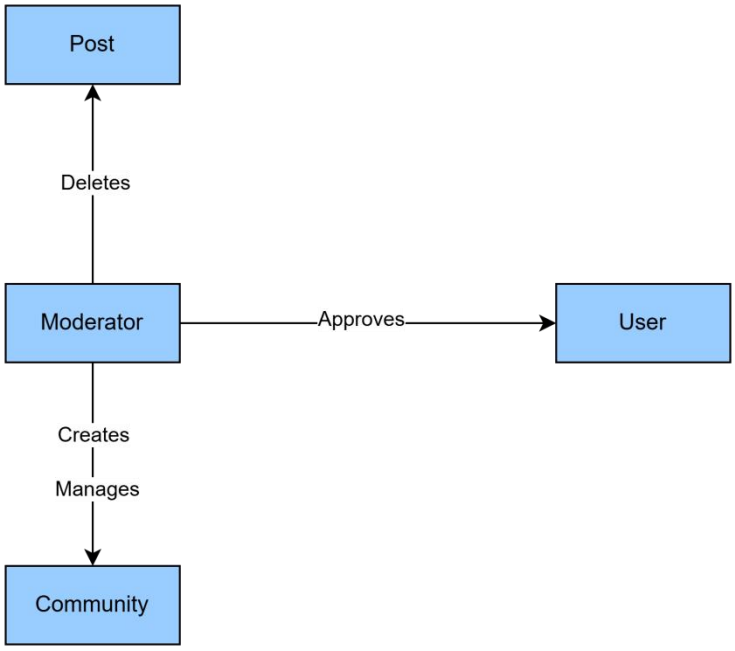


Figure 37. Moderator CRC Diagram

Diagram ID: 13
Name: Gamification

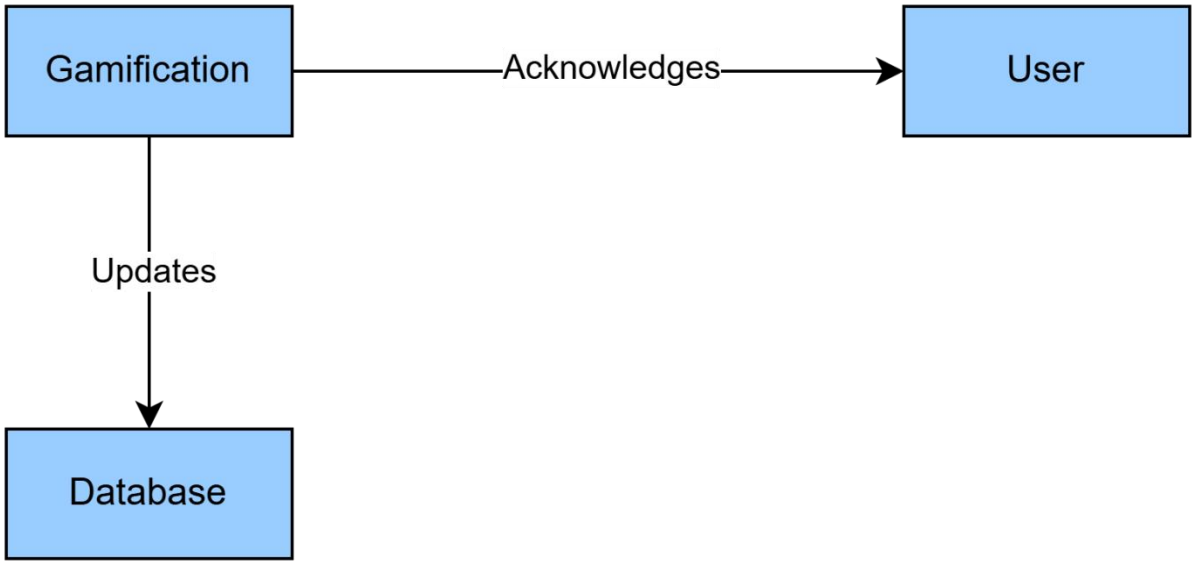


Figure 38. Moderator CRC Diagram

Diagram ID: 14
Name: Driver

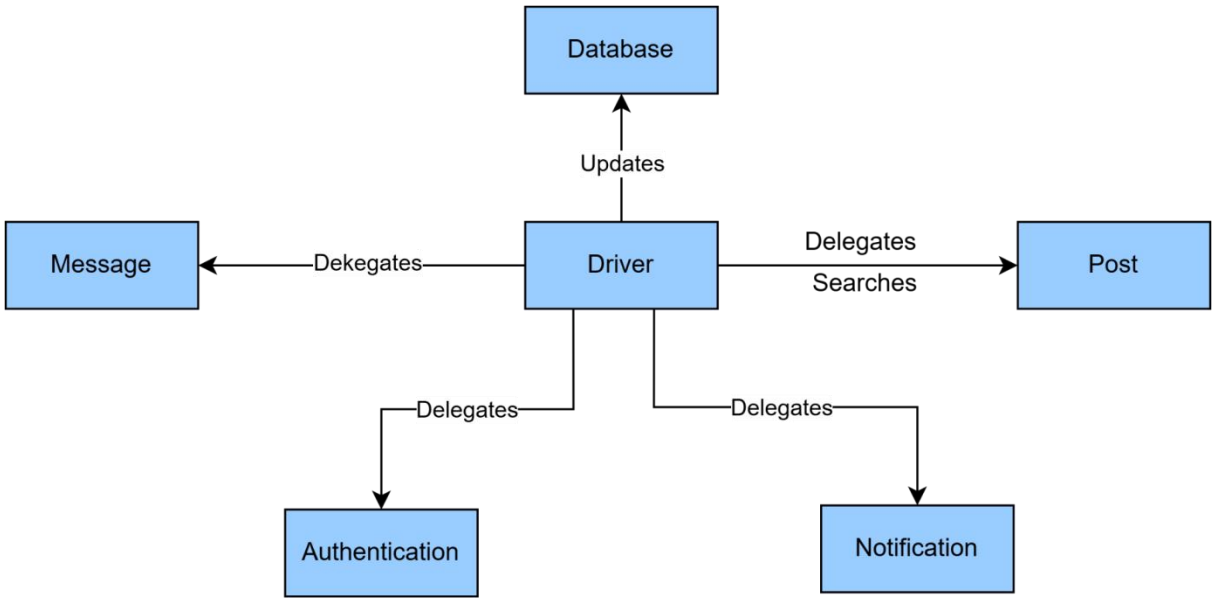


Figure 39. Driver CRC Diagram

Behavioral Modelling

Event Table

Serial	Event	Initiator	Collaborator	Associated Methods
1	Choose to Register	User	Authentication , SendGrid API	+SignUp()
2	Choose to Login	User	Authentication	+Login()
3	Choose to Logout	User		+LogOut()
4	Change Password	User	Authentication	+ChangePassword()

5	Recover Password	User	Authentication, SendGrid API	+ForgotPassword()
6	Update Profile	User		+setCitations() +setHIndex()
7	Create Community	Moderator	Community	+createCommunity()
8	Join Community	User	Community	+joinCommunity()
9	Leave Community	User	Community	+leaveCommunity()
10	Approve	Moderator	Community	+approveMember()
11	Make Post	User	Post	+createPost()
12	Manage Post	User	Post	+editPost() +deletePost()
13	React Post	User	Post	+updatePostVote()
14	Make Comment	User	Post	+makeComment()
15	React Comment	User	Post	+updateCommentVote()
16	Communicate	User	Message, SendGrid API	+sendMessages() +viewMessages()
17	Send Notification	Driver	Notification, SendGrid API	+storeNotification()
18	Summarize	Gemini API	Post	+generateSummary()
19	Update Rewards	Gamification	User	+calculatePoints() +defineBadge()
20	Delete Community	Moderator	Community	+deleteCommunity()

State Transition Diagram

Diagram ID: 1
Name: Authentication

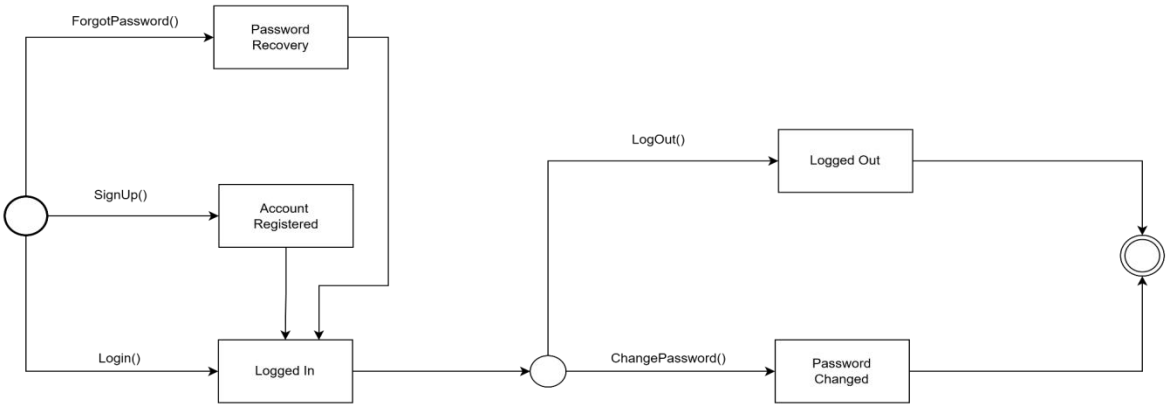


Figure 40. Authentication State Transition Diagram

Diagram ID: 2
Name: User

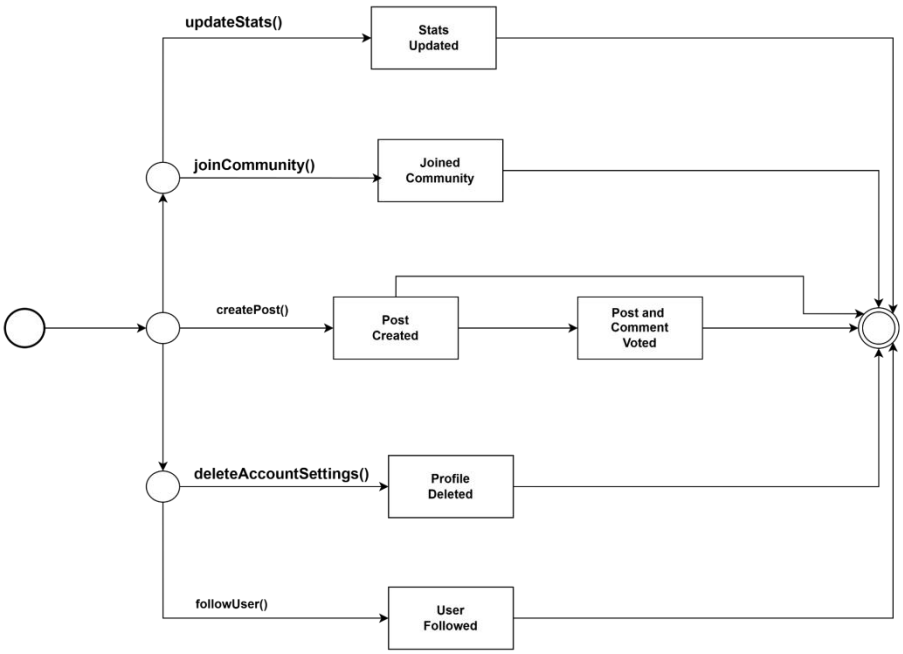


Figure 41. User State Transition Diagram

Diagram ID: 3
Name: Community

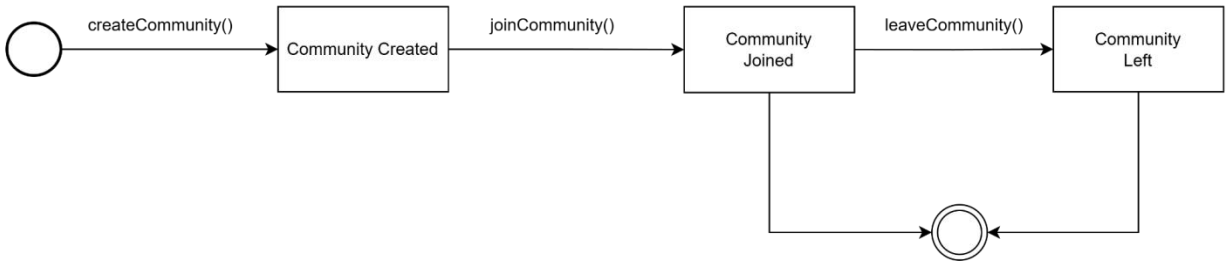


Figure 42. Community State Transition Diagram

Diagram ID: 4
Name: Post

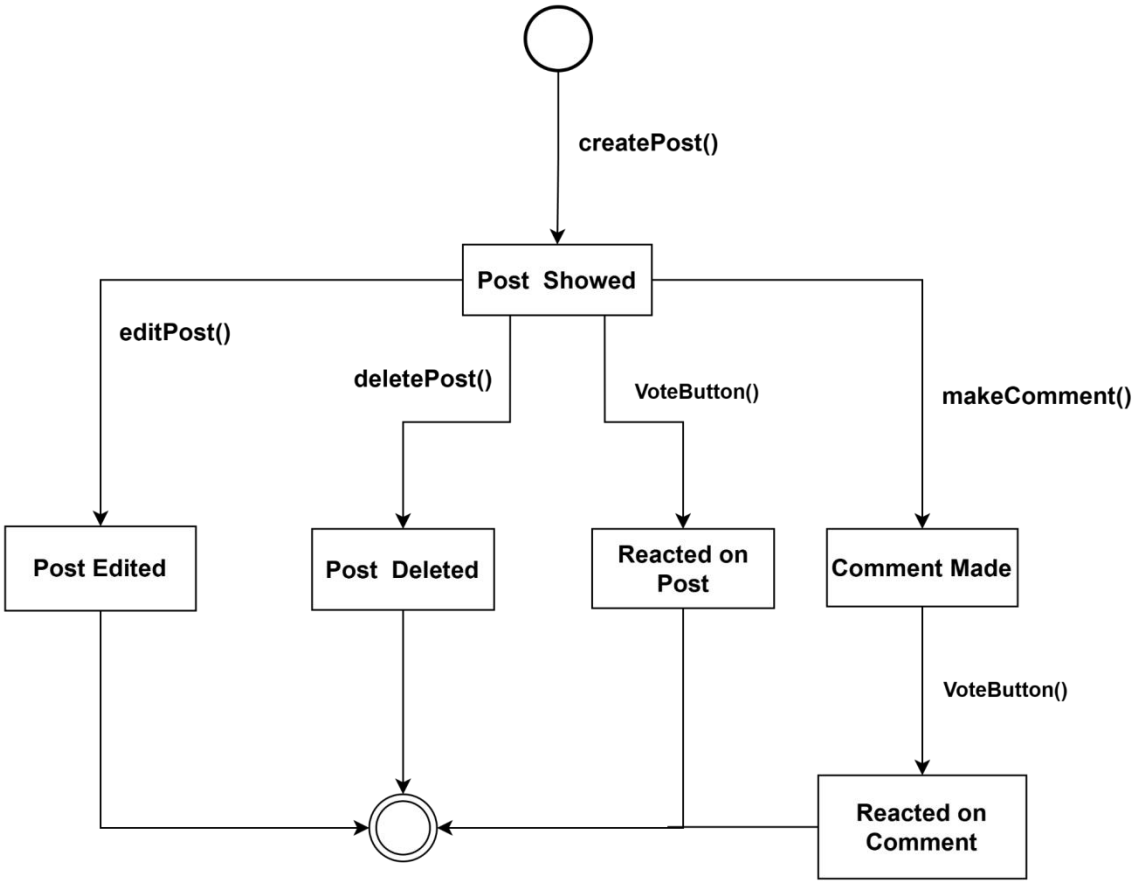


Figure 43. Post State Transition Diagram

Diagram ID: 5
Name: Message

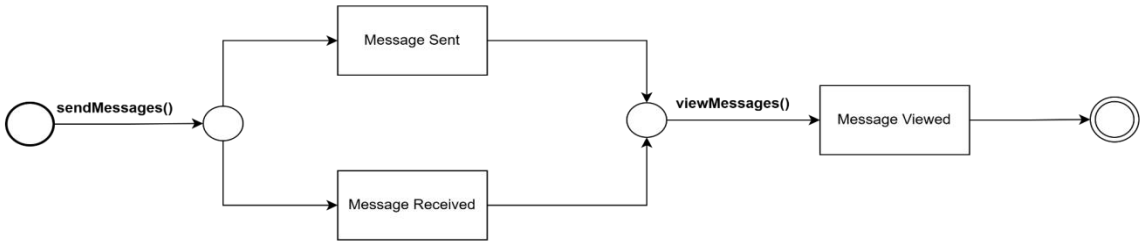


Figure 44. Message State Transition Diagram

Diagram ID: 6
Name: Notification

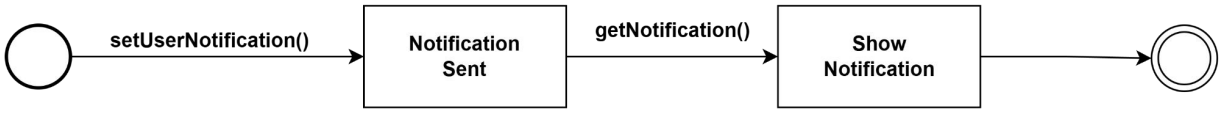


Figure 45. Notification State Transition Diagram

Diagram ID: 7
Name: SendGrid API

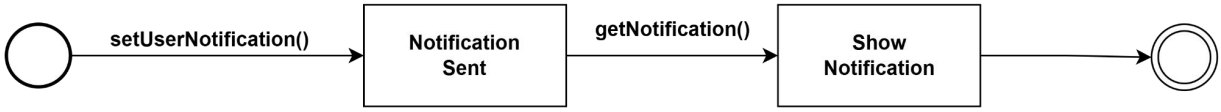


Figure 46. SendGrid API State Transition Diagram

Diagram ID: 8
Name: Gemini API

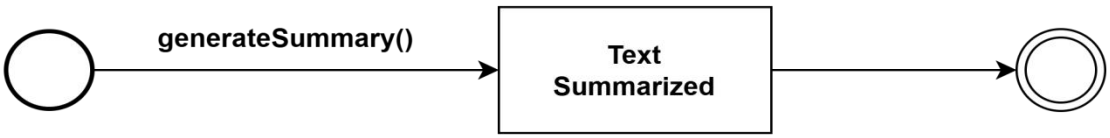


Figure 47. Gemini API State Transition Diagram

Diagram ID: 9
Name: Database

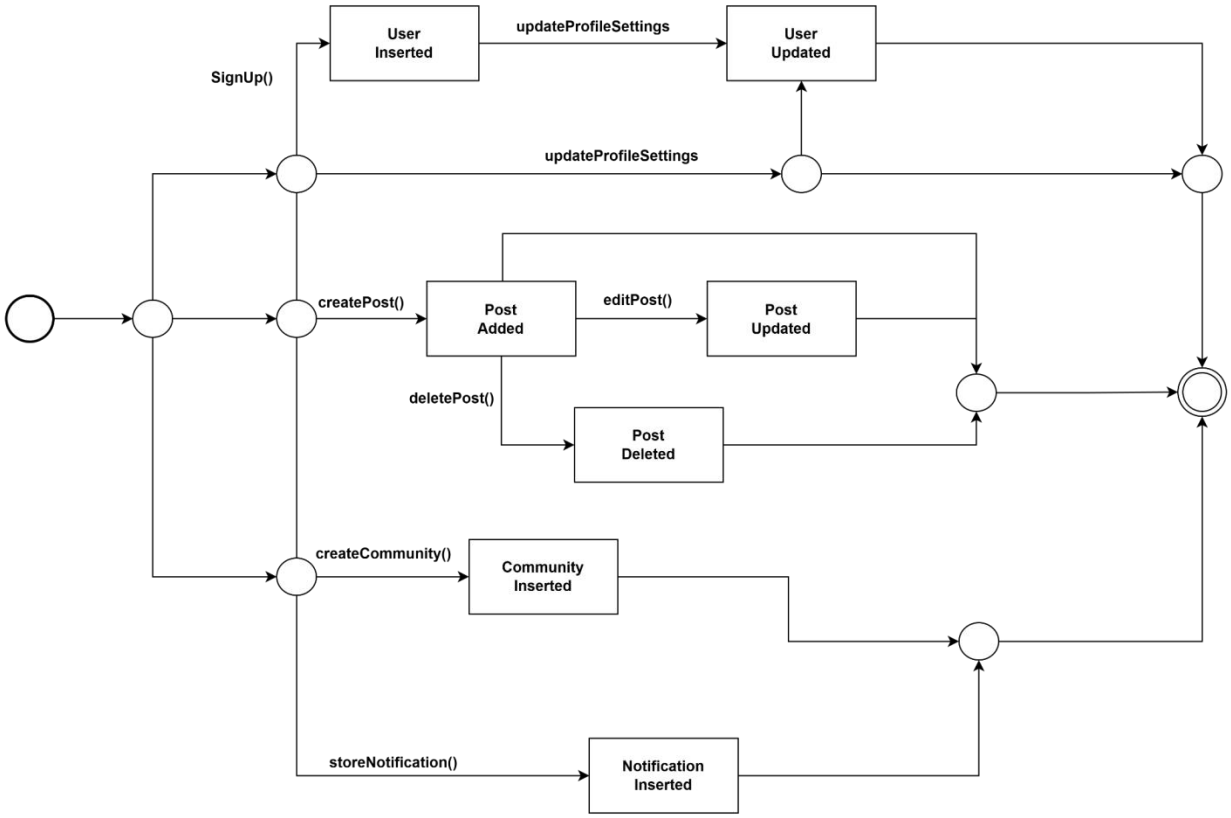


Figure 48. Database State Transition Diagram

Diagram ID: 10
Name: Moderator

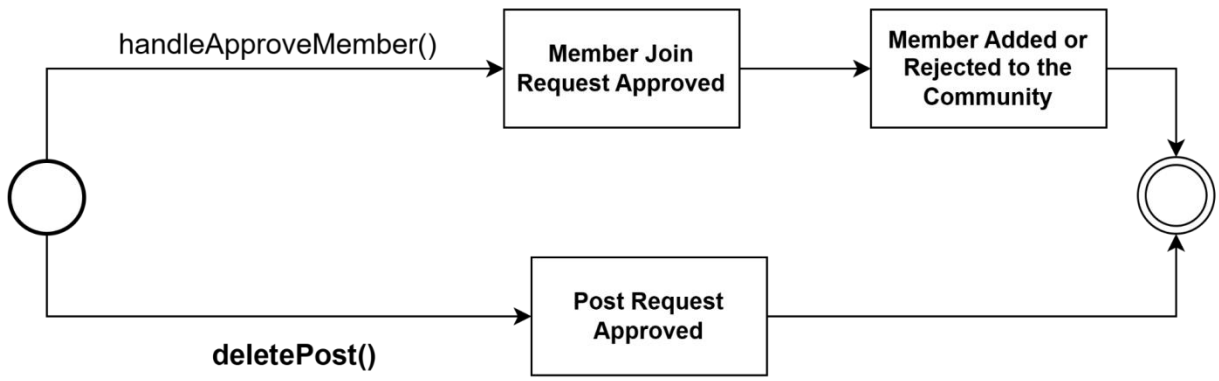


Figure 49. Moderator State Transition Diagram

Diagram ID: 11
Name: Gamification

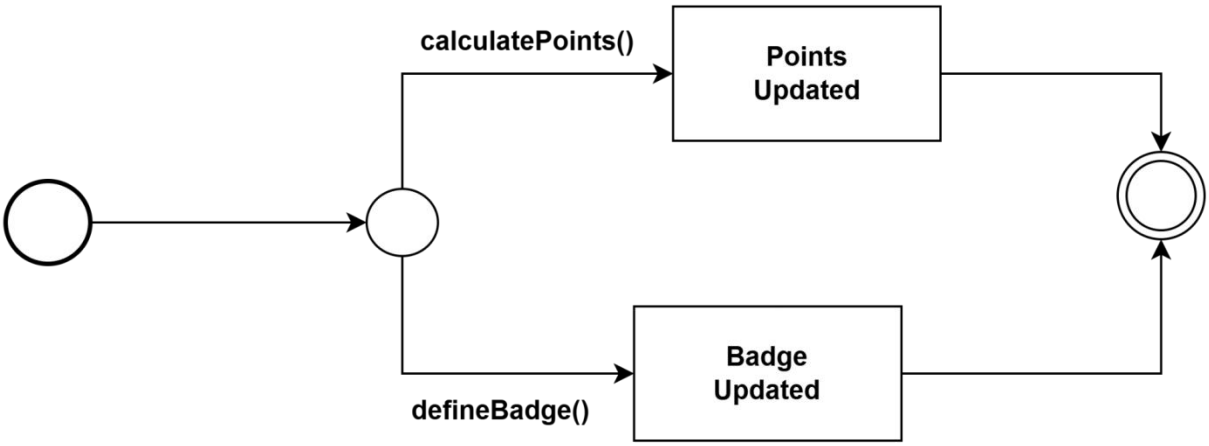


Figure 50. Gamification State Transition Diagram

Diagram ID: 12
Name: Driver

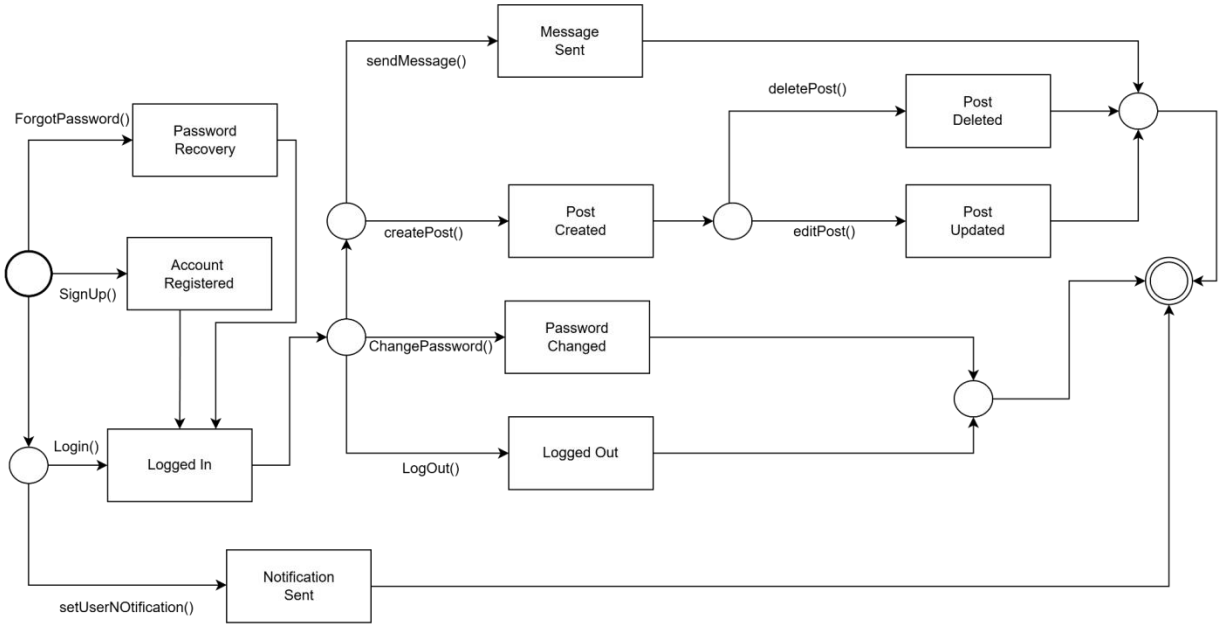


Figure 51. Driver State Transition Diagram

Sequence Diagram

The second type of behavioral representation, called a sequence diagram in UML, is a representation of how events cause flow from one object to another as a function of time. In essence, the sequence diagram is a shorthand version of the use case. It represents key classes and the events that cause behavior to flow from class to class

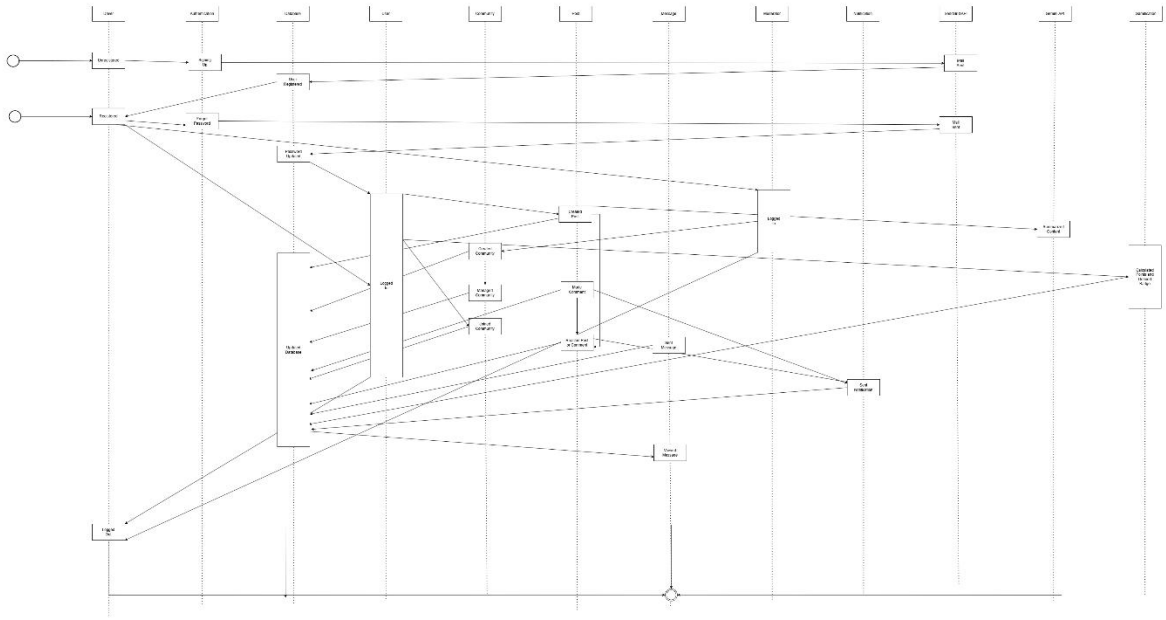


Figure 52. Sequence Diagram

Data Flow Modeling

Level 0

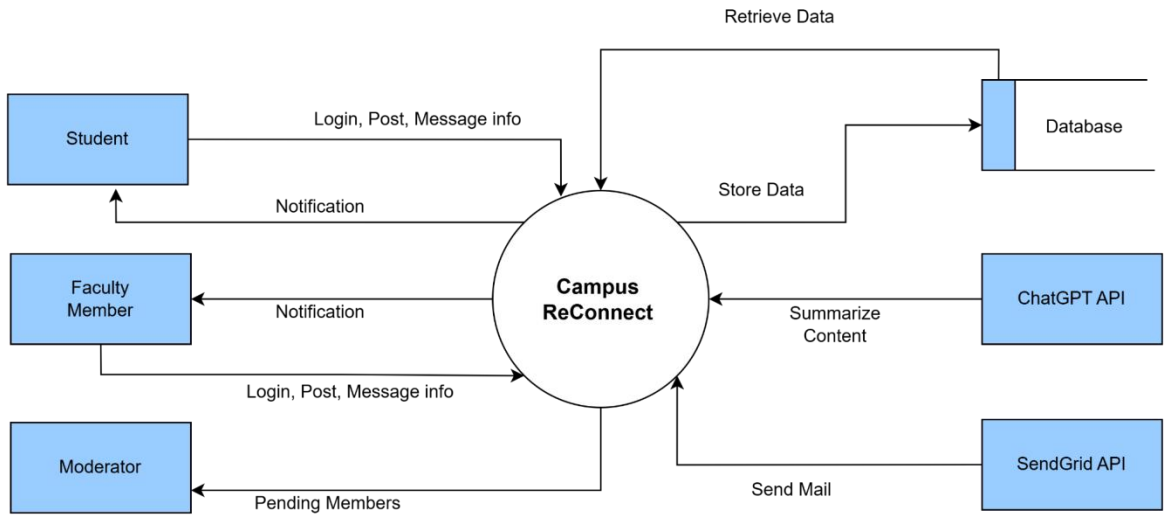


Figure 53. Data Flow Diagram

Level 1

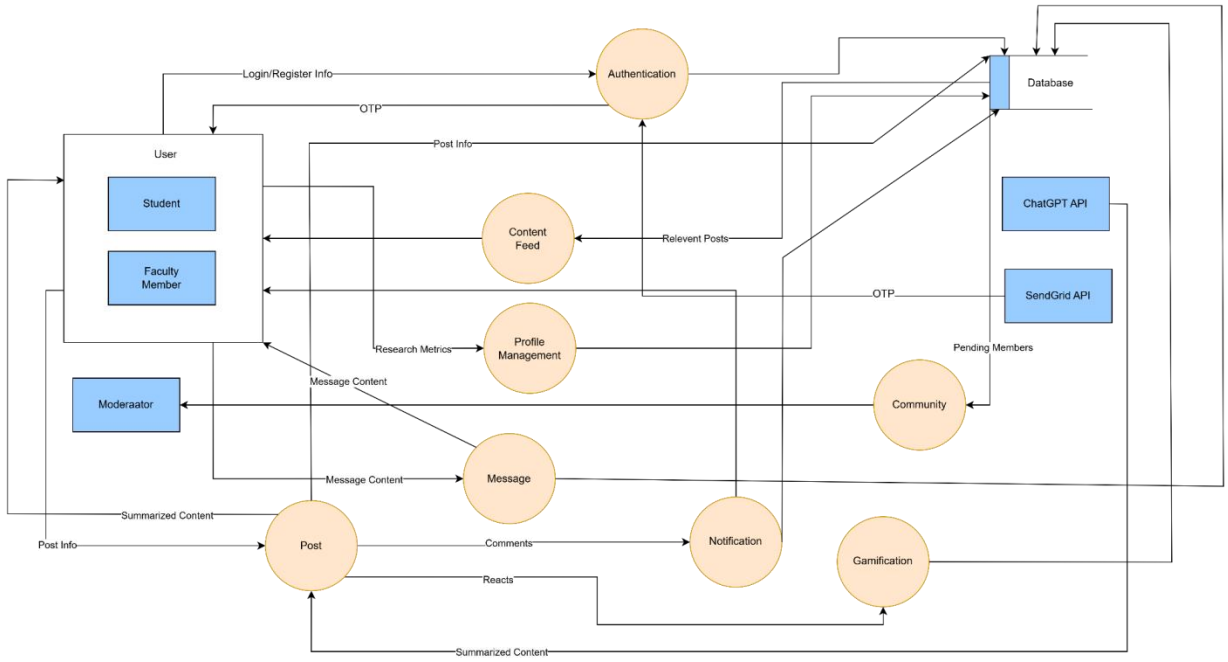


Figure 54. Data Flow Diagram

User Manual

Homepage

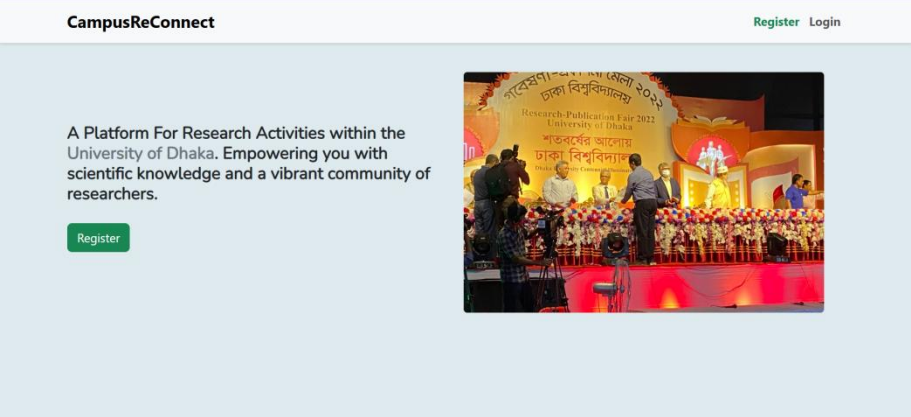


Figure 55. Homepage of the website

Register

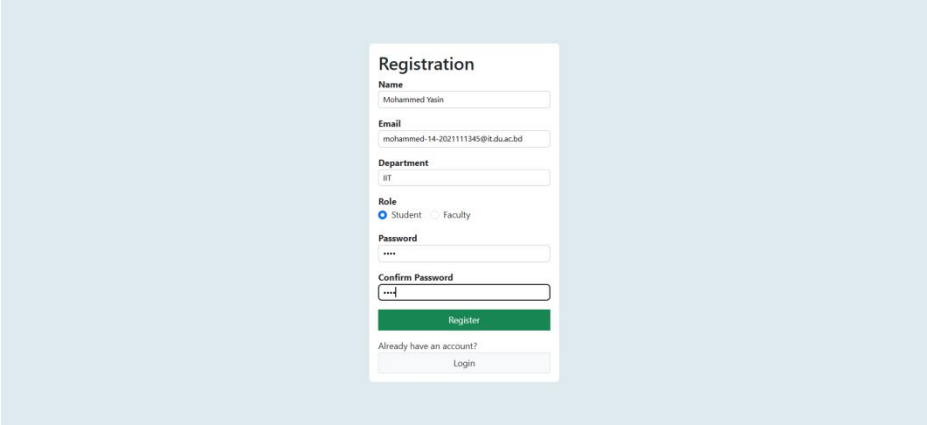


Figure 56. Registration Page

After clicking register users have to register themselves with it.du.ac.bd mail account.

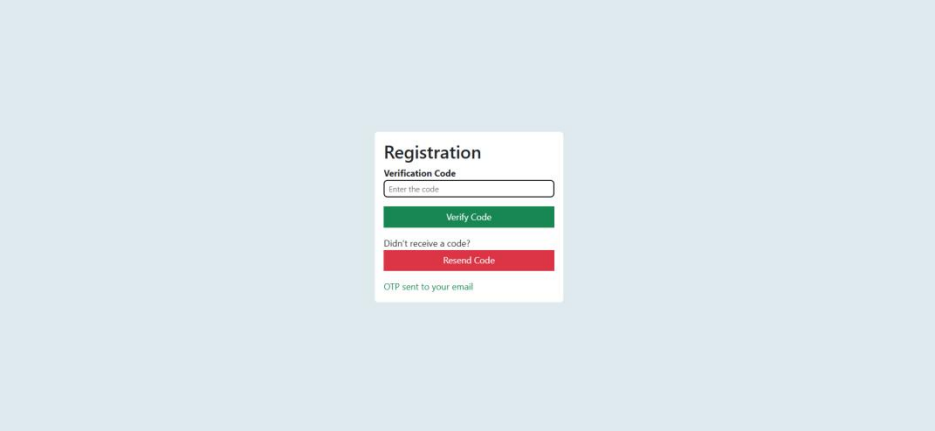


Figure 57. Registration OTP

Users have to confirm an OTP sent to their registered email.

Login

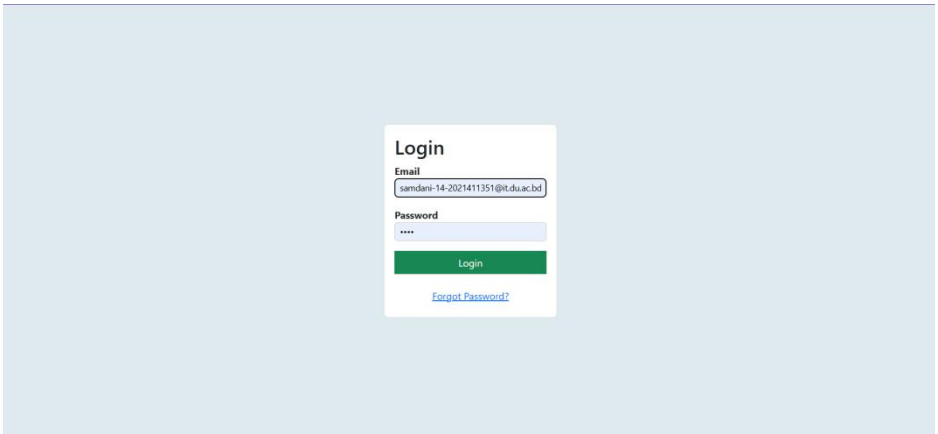


Figure 58. Login Page

After clicking login in the homepage, users will have to provide their credentials in the login page.

Forgot Password



Figure 59. Password Verification OTP

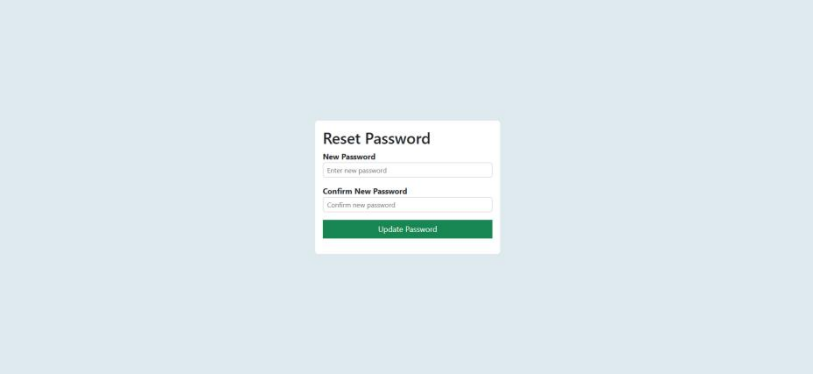


Figure 60. Reset Password

Upon clicking the forgot password in the login page, users will be redirected to the forgot password page where they need to confirm the OTP sent to their email and reset the password.

Feed

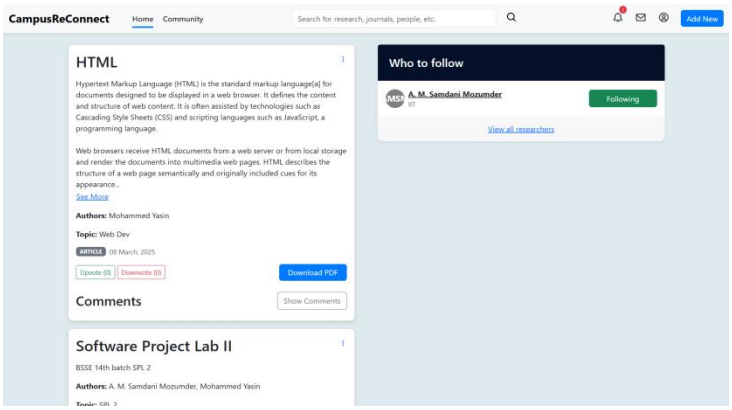


Figure 61. Feed

After logging in, users will be redirected to the content feed. In the content feed, relevant posts will be displayed and on the left some users will be suggested to follow.

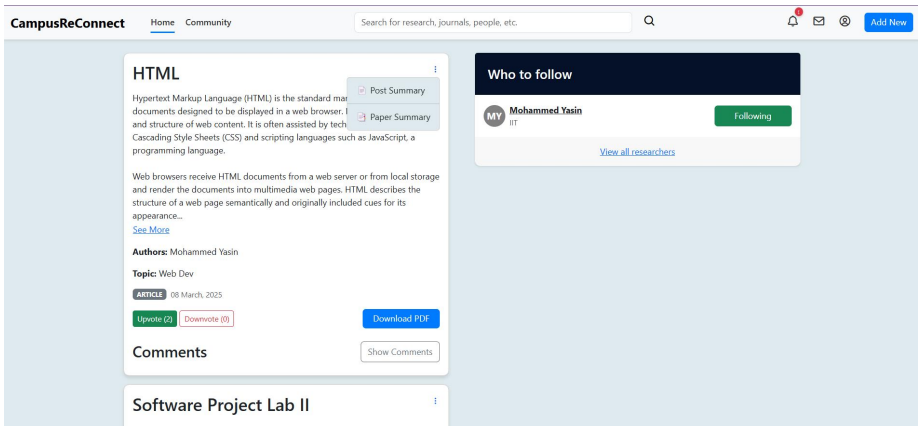


Figure 62. Summary Option

The post description and attached page can be summarized.

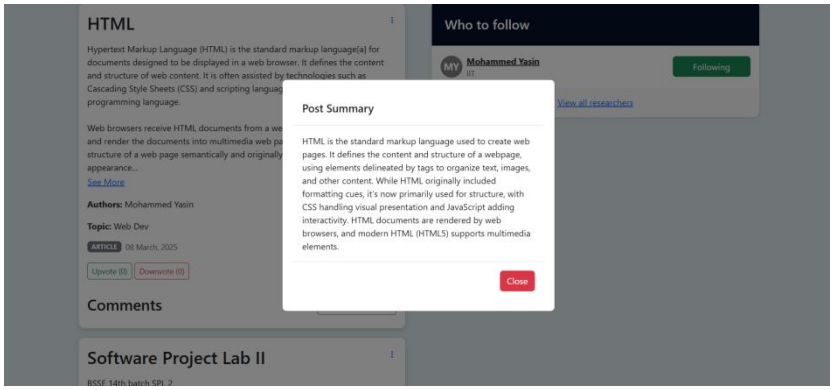


Figure 63. Summarized Content

The summarized content will be shown in a modal.

Profile

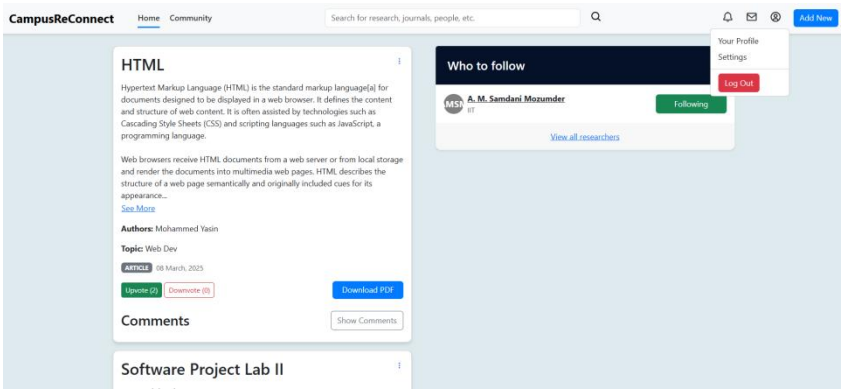


Figure 64. Profile Option

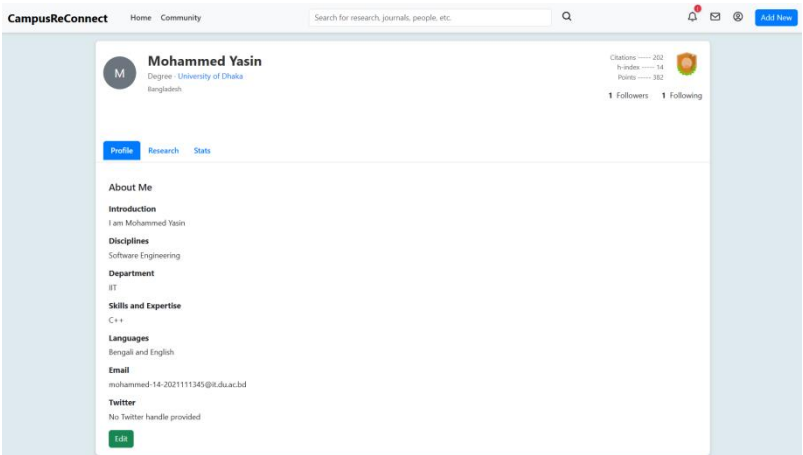


Figure 65. User Profile Tab

The profile icon on the navigation bar has the options for profile tab and account settings. The profile tab shows details of the user. On the top right, there are the research stats, points and the badge of the user.

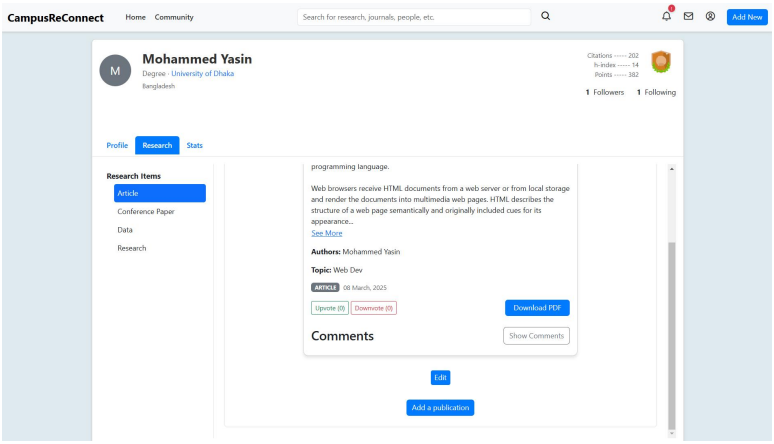


Figure 66. Research Tab

The research tab shows the posts made by the user.

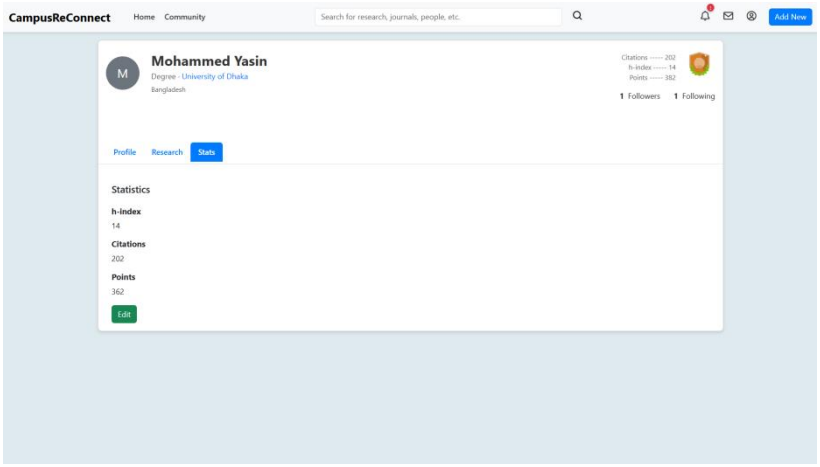


Figure 67. Stats Tab

The stats tab shows the statistics of the user.

Edit Profile

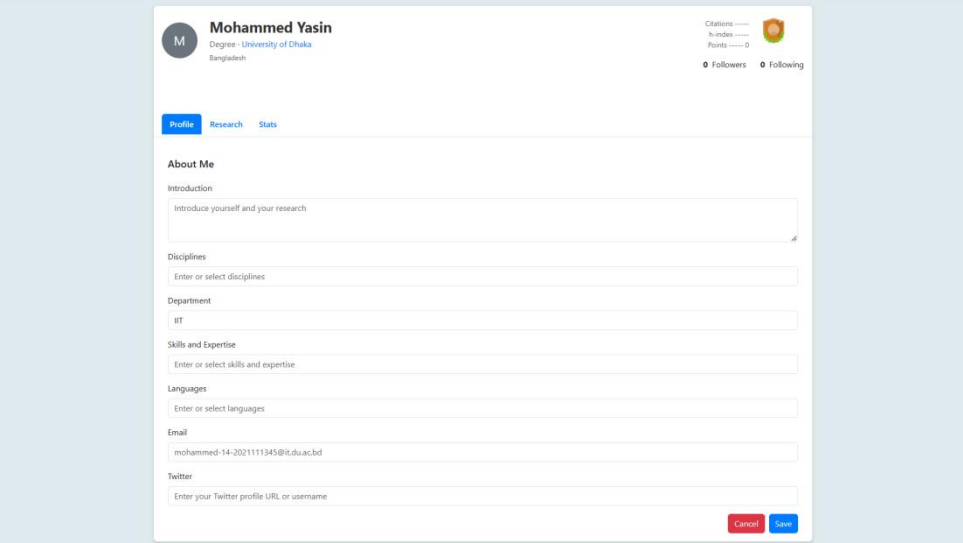


Figure 68. Edit Profile Page

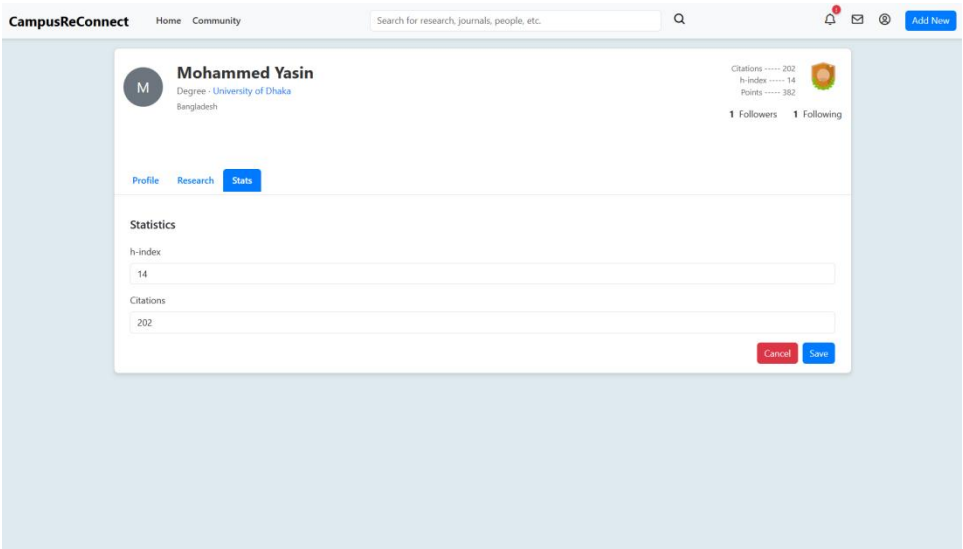


Figure 69. Edit Stats Tab

The profile tab and the stats tab have the edit option to update the fields.

Account Settings

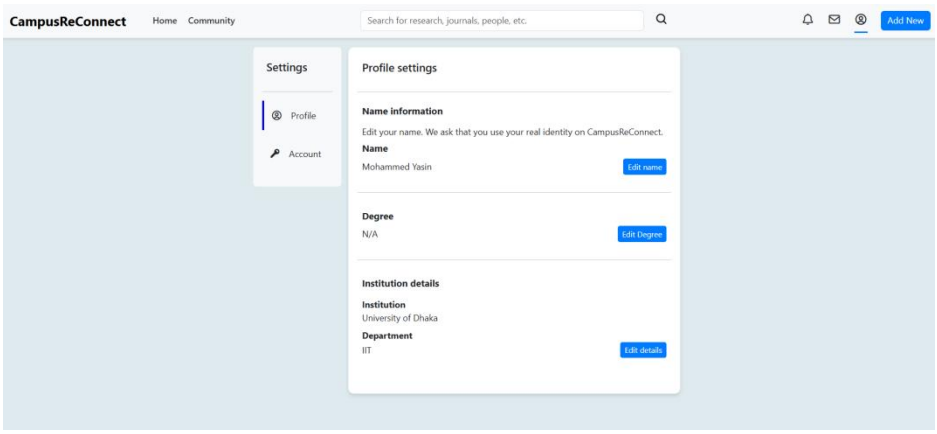


Figure 70. Account Settings

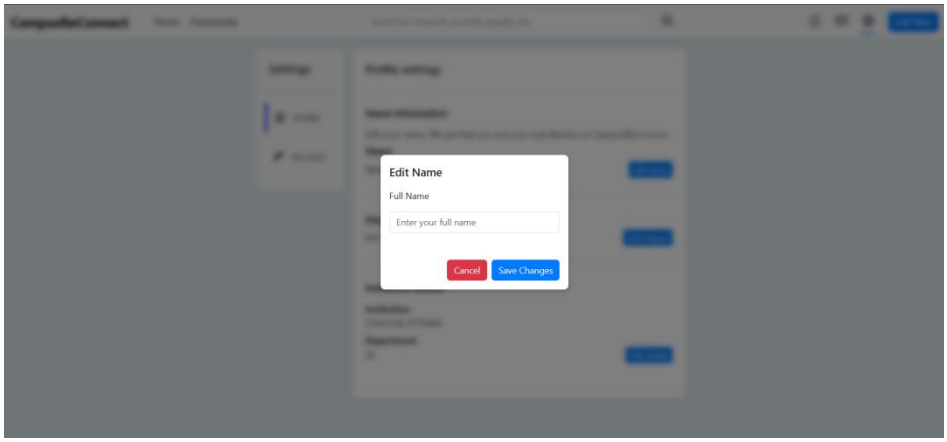


Figure 71. Edit Name

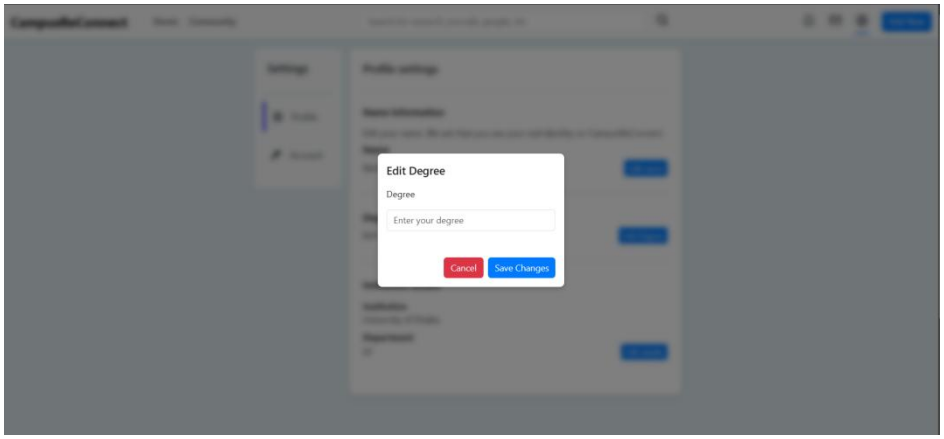


Figure 72. Edit Degree

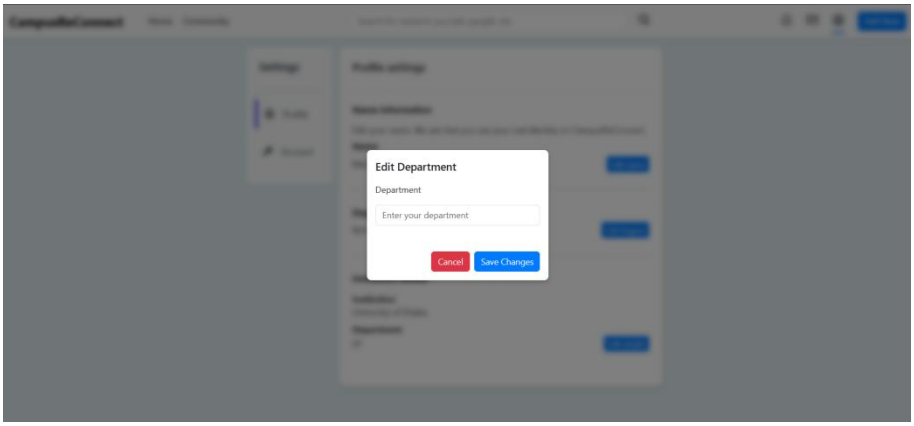


Figure 73. Edit Department

Users can update their information from the account settings option.

Change Password

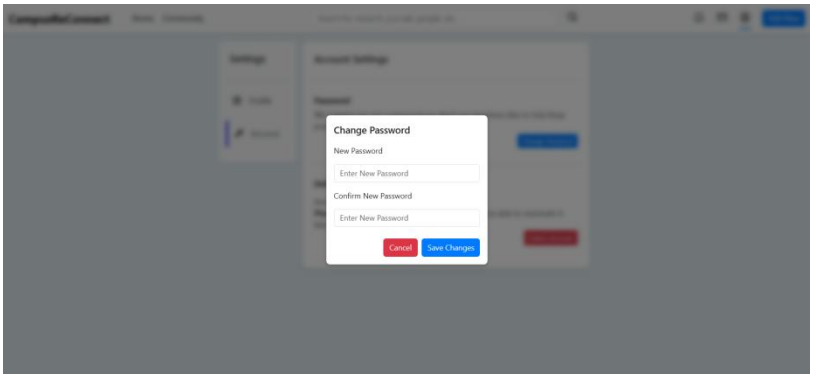


Figure 74. Change Password

Users can change their password by giving the new password and confirming it from the account settings option.

Delete Account

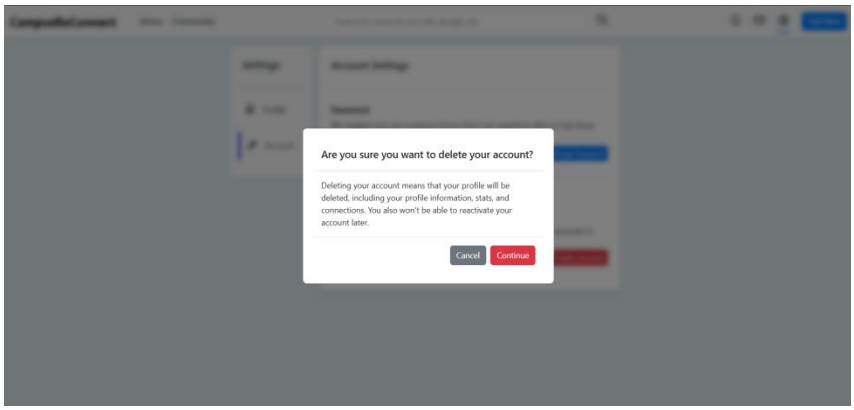


Figure 75. Delete Account

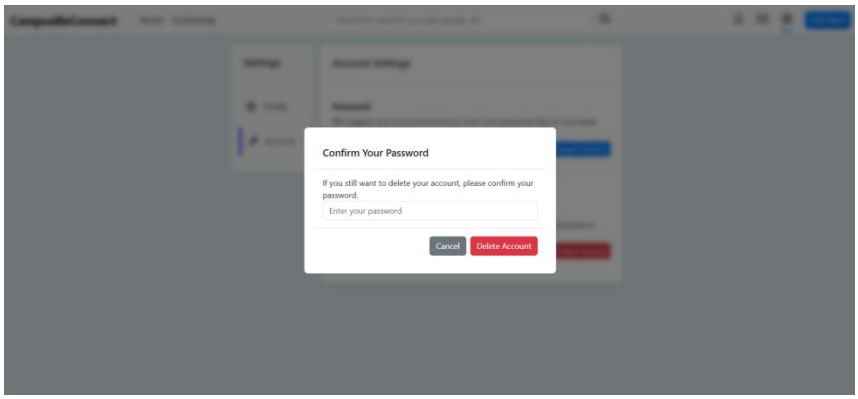


Figure 76. Confirm Password to Delete Account

From the account settings option, users can delete their account after confirming the password.

Create Post

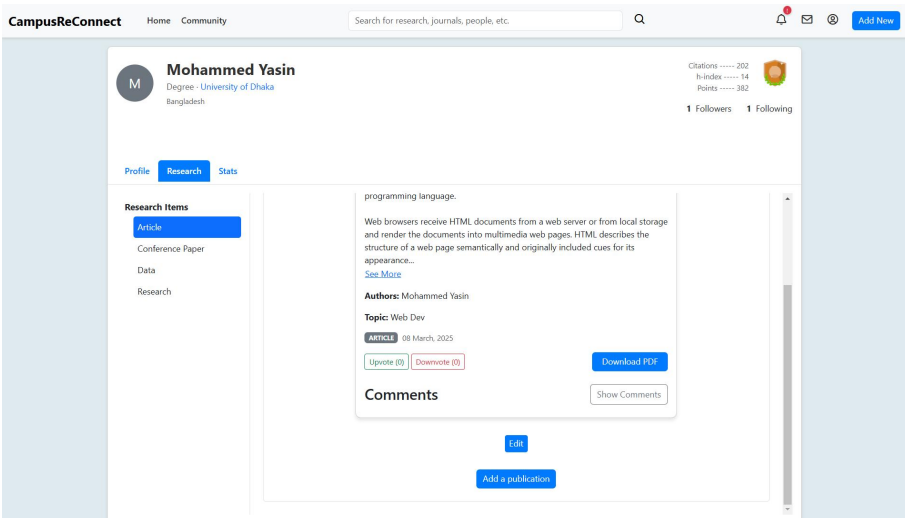


Figure 77. Post Creation

Users can create posts from the research tab of their profile.

Comment and React

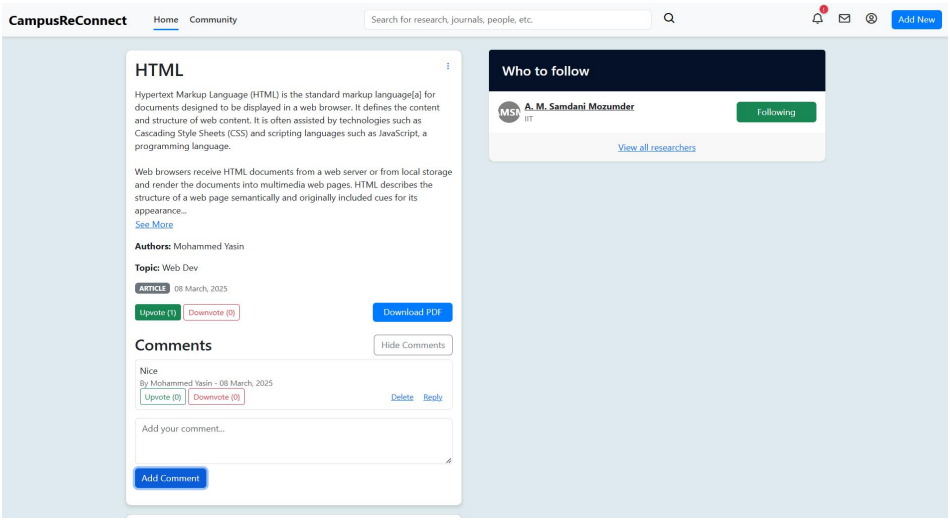


Figure 78. Commenting and Reacting

Users can comment in a post. Votes can be given to the posts as well as the comments.

Message

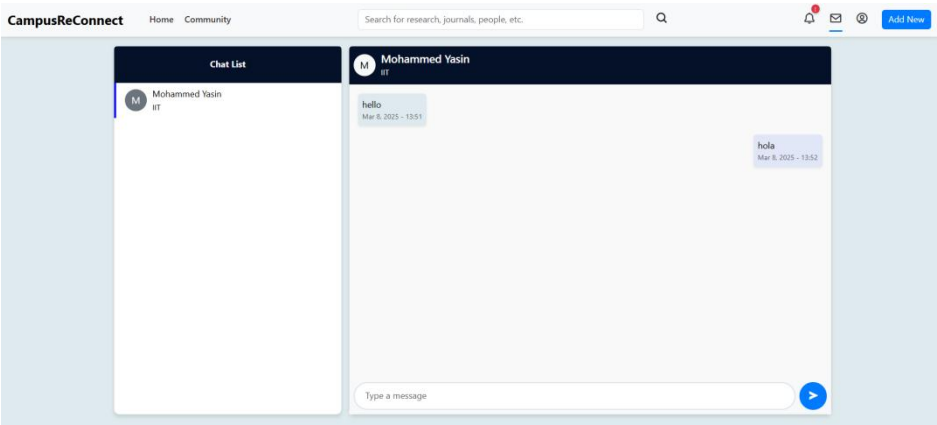


Figure 79. Messaging

Users can send text messages to each other by clicking the message icon on the navigation bar.

Community

Create Community

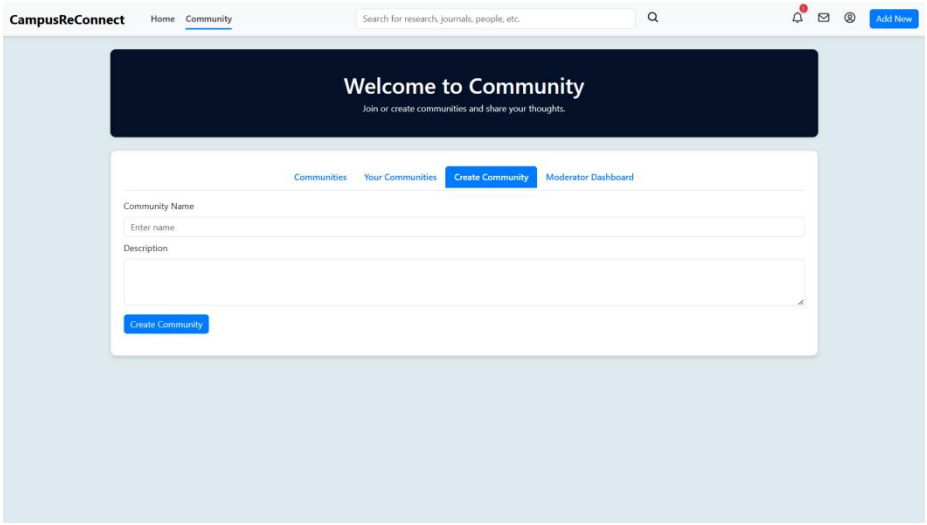


Figure 80. Creating Community

Faculty members can create a community from the Create Community tab in the community page and become moderator of it.

Join Community

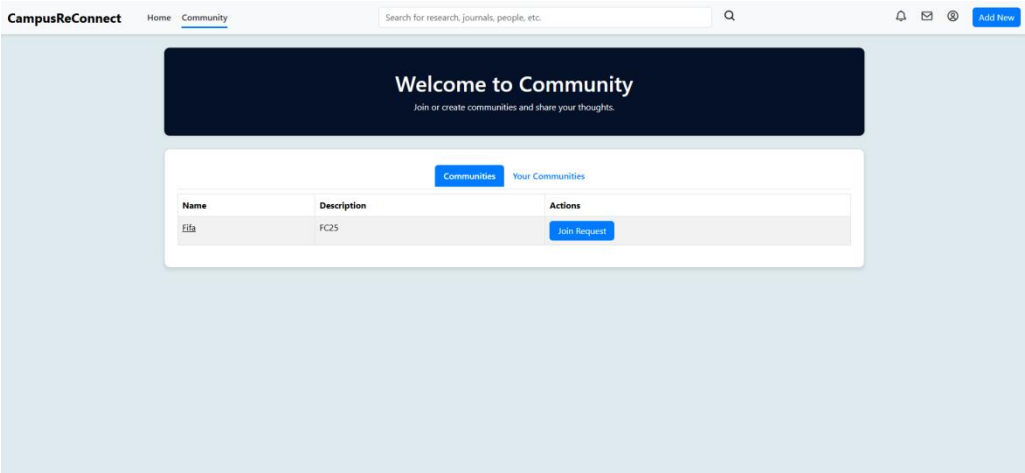


Figure 81. Joining Community

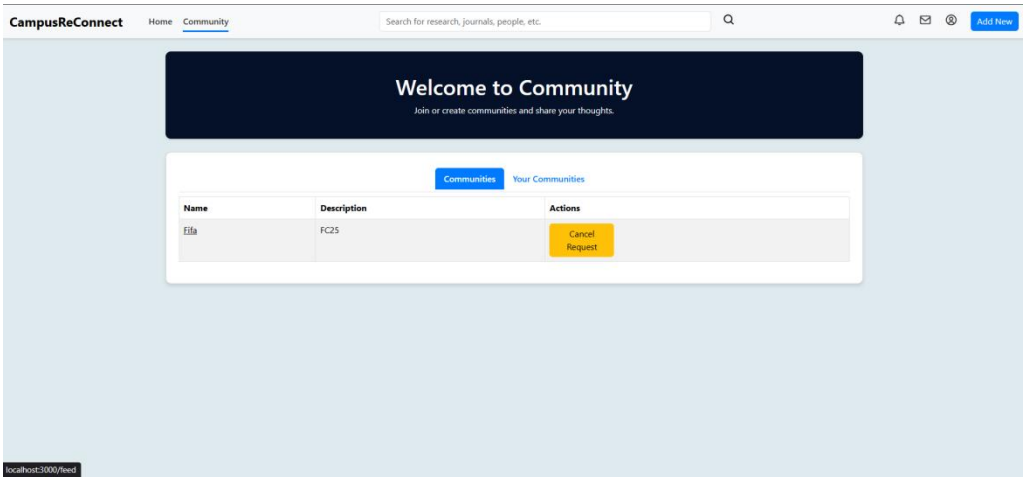


Figure 82. Joining Community (Waiting for Approval)

Users can join a community from the Communities tab in the community page after the approval of the moderator.

Community Feed

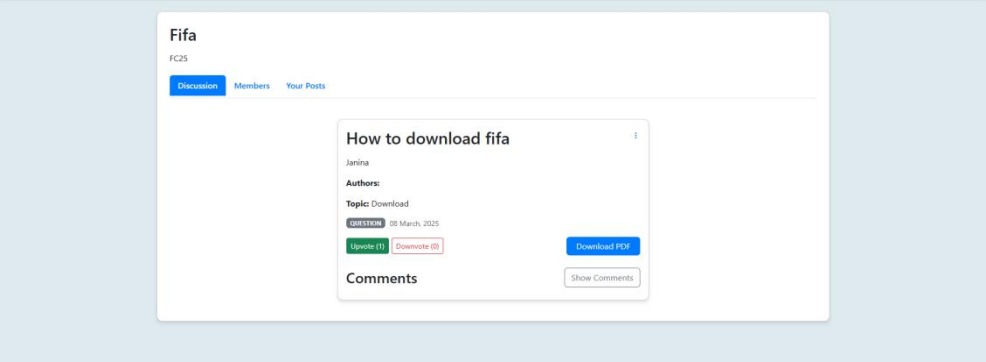


Figure 83. Discussion Tab

Users can see the posts made in the community from the discussion tab of a community.

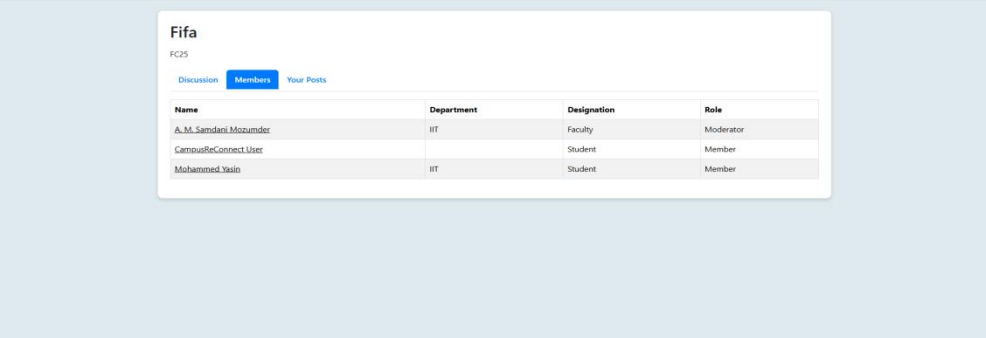


Figure 84. Members Tab

Users can see the members joined in the community from the Members tab of the community.

Post in Community

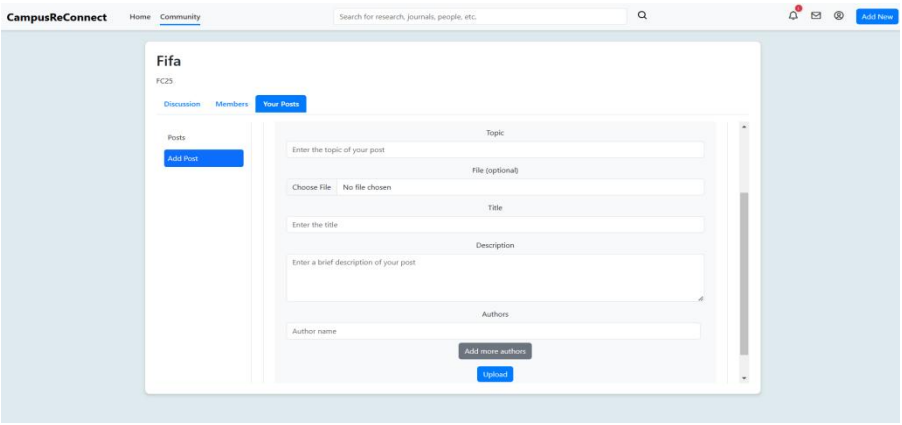


Figure 85. Create Post

Users can post in the community from the Add Post section of the Your Posts tab.

Manage Community

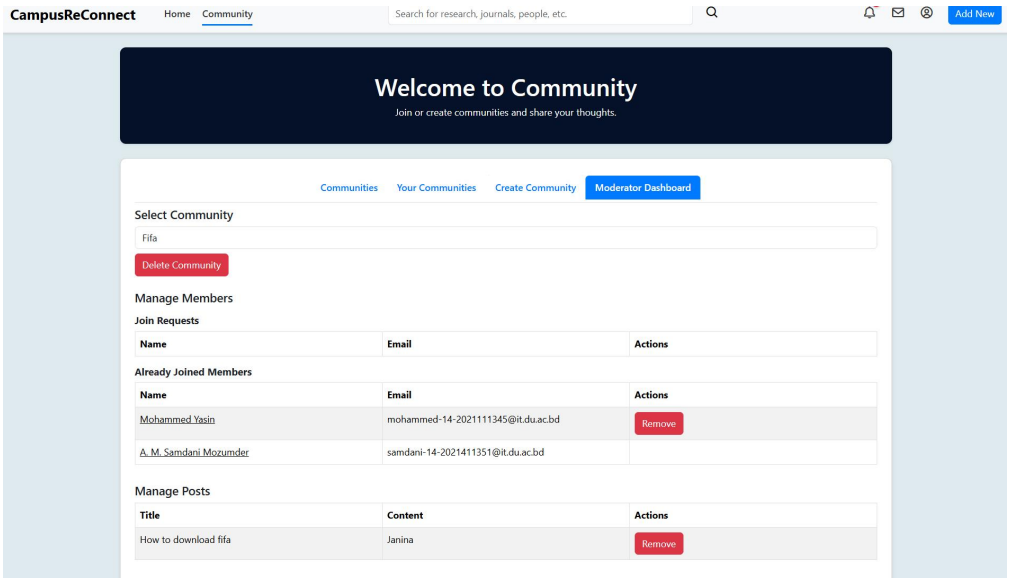


Figure 86. Moderator Dashboard

The moderator can manage the community (removing members, deleting posts and deleting community) from the moderator dashboard.

Notification

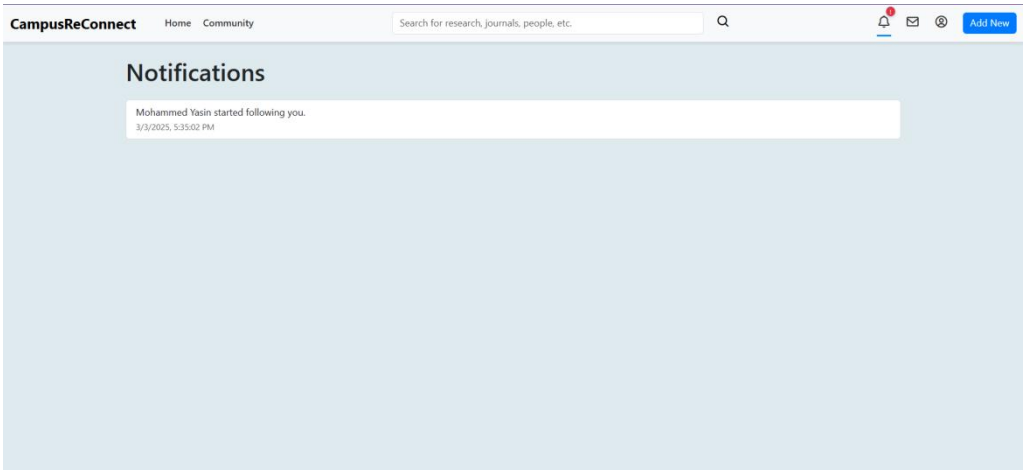


Figure 87. Notification Page

Users can see the notifications sent to them from the bell icon on the navigation bar.

Search

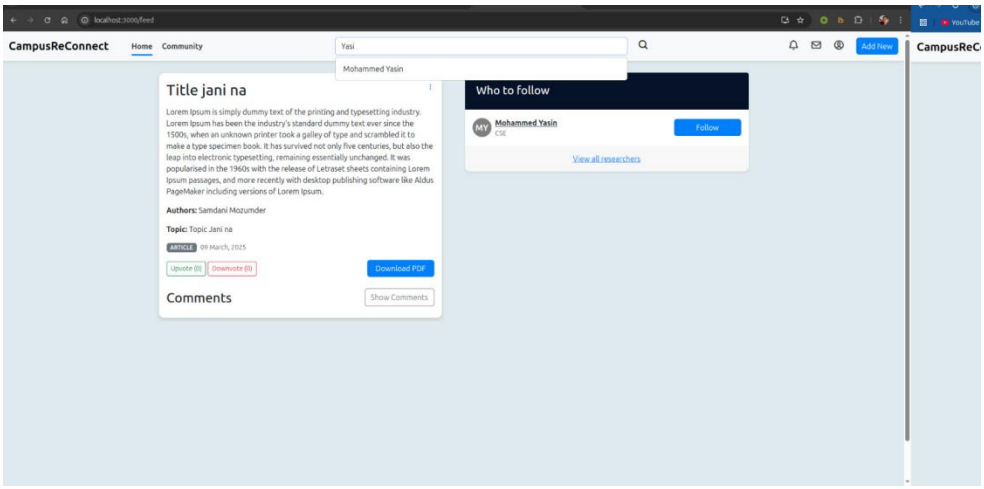


Figure 88. Searching

Users can search other users from the search option on the navigation bar.

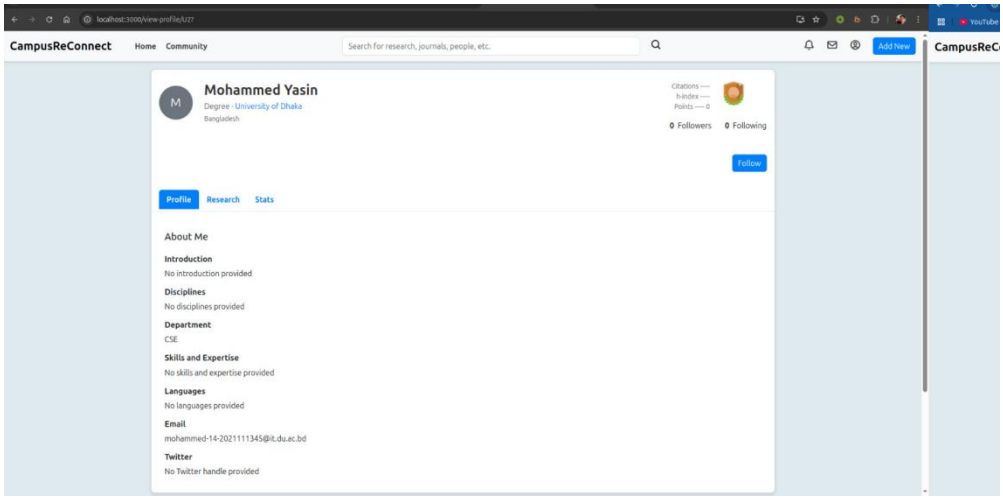


Figure 89. Displaying the Selected Profile

By clicking on the profiles from the search bar, the clicked profile will be displayed.

Challenges Faced

1. Handling such a big project for the first time.
2. Maintaining GitHub repository by multiple people for the first time.
3. Working with socket programming for real time messaging.
4. Handling dependencies with database and the backend.
5. Learning and coding side by side.

Future Work

In the future, we aim to enhance content recommendations by improving the algorithm for more accurate and relevant suggestions based on user interests and activities. Additionally, we plan to introduce support for attachments in messages. If the platform gains significant engagement, we may also explore expanding it beyond the University of Dhaka to create a cross-university platform, fostering collaboration among students and researchers from different institutions.

References

[1] ResearchGate. [Online]. Available: https://www.researchgate.net. (accessed Mar. 9, 2025).